

Smart meter data intelligence for sustainable distribution network operations: State-of-the-Art applications and pathways toward net-zero

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ABSTRACT

The transition toward smarter, more sustainable power systems has positioned data analytics at the core of modern electricity distribution network operations. Enabled by widespread smart meter deployment and advanced sensing technologies, distribution network operators now have access to high-resolution data that supports real-time monitoring, forecasting, and control. This paper presents a comprehensive review of state-of-the-art applications of data analytics in distribution networks, focusing on operational areas such as demand forecasting, electricity theft detection, outage identification, anomaly detection, topology identification, and integration of electric vehicles and energy storage systems. It highlights how advanced techniques, including machine learning, clustering, and deep learning, are being applied to transform raw smart meter data into actionable intelligence. Additionally, the paper discusses the enabling role of digital twins, fog computing, and network intelligence in managing grid complexity, improving system resilience, and supporting decarbonisation goals. Challenges related to data quality, scalability, and interpretability are also explored, emphasizing the need for coordinated technical, regulatory, and institutional responses. The findings underline the critical role of data intelligence in building adaptive, data-driven distribution networks capable of supporting the evolving demands of a low-carbon, decentralized energy future.

1. Introduction

The global energy landscape is undergoing a transformative shift, driven by increasing demand for decarbonisation, decentralization, and digitalization. These transitions are reshaping how electricity is generated, distributed, and consumed, particularly at the distribution level, where emerging technologies and consumer behavior are introducing new operational complexities. In response, distribution network operators (DNOs) are under pressure to enhance grid flexibility, reliability, and resilience while also accommodating distributed energy resources (DERs), electric vehicles (EVs), and dynamic demand-side participation. The successful management of this evolving system relies heavily on the ability to extract timely and actionable insights from large volumes of diverse, high-frequency data.

Smart meters have emerged as foundational tools in this digital transformation. Their widespread deployment has created new opportunities for near real-time monitoring and advanced analytics in

distribution networks. By continuously recording detailed measurements of voltage, current, power factor, and energy consumption, smart meters enable DNOs to observe system conditions at granular spatial and temporal scales [1]. This visibility facilitates the shift from traditional, centralized grid management to a more decentralized and data-driven operational model. As of 2020, over 100 million smart meters had been deployed in Europe alone, with forecasts indicating continued growth globally [2].

However, data availability alone is insufficient. The true value of smart meter infrastructure lies in how effectively the data is utilized. This requires integrating big data analytics techniques, such as machine learning, statistical modeling, clustering, and deep learning, into operational workflows. These techniques can be used to address a wide range of challenges, from demand forecasting and load profiling to outage detection, theft identification, and network topology inference. When appropriately applied, data analytics not only improve situational awareness but also support automated decision-making, thereby

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enhancing the responsiveness and efficiency of distribution network operations [3,4].

Despite its potential, the application of big data in electricity distribution networks presents significant challenges. Raw data collected from smart meters often suffers from quality issues such as noise, missing values, and inconsistent timestamps, necessitating robust pre-processing and feature selection techniques [5]. Moreover, machine learning models must be carefully tailored to account for the non-linear, context-specific nature of electricity usage and grid behavior. The risk of overfitting, lack of interpretability, and computational demands also pose barriers to the effective integration of analytics tools into existing DNO systems. The journey of data from smart meter devices, collected and processed through analytics systems, to serve various stakeholders such as distributors, consumers, policymakers, and retailers is depicted in Fig. 1. The insights derived from smart meter data analytics enable applications in network intelligence, including demand forecasting, network planning, theft detection, outage detection, anomaly detection, and phase and topology identification.

Another layer of complexity arises from the increasing penetration of DERs and electrification of end-use sectors, particularly transport. These changes introduce highly dynamic, bidirectional power flows and shift load patterns in ways that traditional distribution planning and operation frameworks are not designed to manage. The integration of EVs, rooftop solar, and home energy storage requires DNOs to implement intelligent coordination mechanisms supported by real-time data streams and predictive algorithms [6]. Without adequate analytics capabilities, these new technologies could introduce instability and operational uncertainty rather than enhancing grid flexibility.

At the same time, DNOs face considerable challenges in harnessing the full potential of smart meter data, particularly in the real-time processing and analysis of large, high-frequency datasets. Addressing these challenges requires the development of innovative, scalable, and practical analytical methods that can extract actionable insights to support network planning, operational decision-making, and enhanced grid observability [7]. Given the significant value embedded in smart meter data, identifying and deploying appropriate analytics techniques is essential to maximize its impact on the efficiency, reliability, and adaptability of modern distribution systems. This paper presents a comprehensive review of state-of-the-art applications of big data analytics in support of electricity distribution network operation. It examines how data collected from smart meters is used to address key operational challenges and optimize emerging use cases. The paper begins by characterizing smart meter data and pre-processing methods, then surveys advanced analytics applications across several operational domains including forecasting, anomaly detection, topology identification, outage response, and integration of EVs and storage systems. It concludes by discussing systemic issues, future directions, and the enabling role of digital technologies and network intelligence in transitioning toward sustainable, data-driven power distribution systems.

2. Smart meter data analytics

Smart meter data serves as the cornerstone of data-driven intelligence in modern distribution networks. This section establishes the technical groundwork by detailing the types, formats, and temporal characteristics of data generated by smart meters, and the crucial role of pre-processing in ensuring its usability. It highlights how the quality, resolution, and organization of this data directly influence the performance of analytics tools such as forecasting, clustering, and anomaly detection. Section 2 provides the essential context for understanding how raw data is transformed into actionable insights that support advanced grid operations.

2.1. Types and characteristics of smart meter data

Smart meters generate a diverse range of measurements that provide detailed visibility into how electricity is consumed, delivered, and experienced across households, buildings, and distribution networks. These data streams differ widely in temporal granularity, measurement precision, informational richness, and operational purpose. Understanding these variations is essential for selecting appropriate analytical methods and for designing scalable, reliable data-driven applications. This section provides an expanded perspective of smart meter data types, structured to highlight the technical characteristics that influence their analytical value.

The most common category is temporal consumption data, which records active and reactive energy at regular time intervals. These data form the backbone of many analytical tasks because they capture fundamental load patterns that reflect user activity, appliance operation, and environmental conditions. Depending on the technological maturity of the metering infrastructure, sampling intervals may range from hourly readings to sub-second measurements. Higher-frequency data provide improved visibility of transient load variations and appliance signatures, while lower-frequency readings are typically used for long-term modeling or strategic planning. Although temporal consumption data are relatively simple in structure, their analytical performance is sensitive to resolution, signal stability, and the presence of missing or corrupted intervals. As a result, the usefulness of these datasets is tightly linked to the characteristics of the metering device and the quality of network communication channels.

A second category consists of electrical parameter measurements, including voltage, current, frequency, and power factor. These measurements offer a deeper view into the physical behavior of residential and commercial circuits and provide essential evidence for assessing power quality and network stability. For instance, voltage deviations can indicate local congestion, poor transformer tap settings, or stresses caused by the integration of DERs. Similarly, current waveforms and harmonic content can reveal abnormal equipment operation or the presence of non-linear loads. These data are typically produced at higher sampling frequencies than basic consumption data, since power quality



Fig. 1. Flow of smart meter data from collection to stakeholder insights, enabling key applications in network intelligence such as forecasting, outage detection, and topology identification.

issues evolve rapidly and require timely detection. From an analytical standpoint, electrical parameters enrich model inputs by introducing operational context, though they also increase complexity and computational demand.

Smart meters also produce event driven data, which are generated only when specific conditions occur. Examples include outage notifications, voltage sag and swell events, breaker trips, tamper alerts, and restoration timestamps. These data have substantial operational value because they indicate sudden changes in network state that require timely intervention. Event driven data are not continuous time series. Instead, they capture key moments that disrupt normal system performance. Their analytical significance lies in their ability to validate anomaly detection algorithms, support reliability assessments, and assist in fault location processes. The primary challenges with event driven data are inconsistent timestamp accuracy, variation in reporting formats across device manufacturers, and potential data loss when communication channels are constrained during system disturbances.

As modern energy systems become more decentralized, smart meters increasingly operate alongside devices that generate cross domain energy data. These include electric vehicle chargers, solar inverters, battery storage units, heat pumps, and smart HVAC systems. Each subsystem introduces its own data structure, sampling rate, and measurement semantics. For example, photovoltaic inverters may report panel output every few seconds, while EVs may provide charging rate, battery status, and connection events. When combined with traditional meter readings, cross domain datasets enable multi vector analysis such as flexibility estimation, co-optimization of local resources, and coordinated control strategies. However, these datasets also introduce heterogeneity, which necessitates additional preprocessing and alignment to ensure interoperability.

The last category is metadata and contextual information. Although these data do not represent electrical measurements, they provide essential context that improves the performance and interpretability of data-driven models. Examples include tariff structures, weather data, occupancy schedules, building insulation characteristics, appliance ownership, and socio-demographic attributes. These data help explain variation in consumption patterns and can significantly enhance model accuracy, especially for personalised or behaviour-informed applications. Metadata often require careful validation because inconsistencies or outdated attributes can create misleading analytical outcomes.

Table 1 summarizes these data categories and highlights the main characteristics that differentiate them. The table illustrates how each type aligns with specific analytical tasks and shows the typical resolution range and measurement purpose. In practice, effective smart meter analytics rely on combining multiple data types to create a comprehensive representation of energy use and system conditions. For instance, temporal consumption data alone may capture overall demand, but integrating power quality measurements or contextual metadata can refine forecasts and improve anomaly detection accuracy. Similarly, incorporating event driven data and cross domain information supports advanced capabilities such as segmentation of flexible loads or detection of interactions between distributed resources and local

demand.

Overall, the diversity of available smart meter data allows researchers and practitioners to design advanced analytical frameworks capable of supporting real time decision making, operational optimization, and strategic planning. A detailed understanding of these data types provides the foundation for selecting suitable algorithms, managing data quality issues, and developing robust models that account for the complex behavior of modern distribution networks.

2.2. Data pre-processing

Data pre-processing is a foundational step in any data-driven application within distribution network operations. Smart meter data, while rich in information, is susceptible to noise, gaps, and anomalies that can undermine the effectiveness of analytical models. Pre-processing addresses these issues by transforming raw data into a structured, clean, and reliable format that is suitable for input into forecasting models, classification algorithms, or clustering tools. Basic pre-processing tasks include handling missing values, smoothing noisy signals, normalizing data ranges, and removing random outliers that can skew model accuracy. These steps are particularly important in time-series analysis, where even minor inconsistencies in input data can lead to significant forecasting errors or misclassifications.

Beyond basic cleaning, advanced data processing techniques are required to tailor datasets to the needs of specific analytical tasks. For instance, when attempting to forecast electricity demand for thousands of individual customers, treating each consumer independently is often computationally inefficient and prone to data sparsity issues. A more scalable and effective approach involves grouping consumers with similar load characteristics and forecasting at the cluster level. This approach not only improves model efficiency but also reveals consumption subgroups with distinctive or abnormal usage patterns, offering valuable insights into behavioral trends or potential anomalies in the network [8].

One of the most widely used techniques in this context is clustering, which segments electricity consumers based on similarities in their energy use profiles, demographic characteristics, or geographic proximity. These clusters can be used to develop more targeted demand response strategies or to identify customers with irregular load behaviors. Methods such as k-means clustering, k-nearest neighbors (KNN), and self-organizing maps are commonly applied to smart meter datasets for this purpose [9]. For example, Haben et al. [10] demonstrated the use of a finite mixture of Gaussian distributions to divide residential load profiles into meaningful clusters, serving as a precursor to demand-side management interventions. Such segmentation can guide utilities in customizing tariffs, identifying high-load regions, or prioritizing customer engagement efforts.

Clustering also supports analysis of energy technology impacts. For instance, it has been applied to evaluate the influence of battery storage systems and rooftop PV on household energy consumption dynamics. By comparing clustered consumption patterns before and after technology adoption, researchers can identify trends that inform both regulatory

Table 1
Smart meter data types, resolution, fidelity issues, and applications.

Data Type	Example Parameters	Typical Resolution	Fidelity Considerations	Typical Analytics Use Cases
Temporal consumption data	Active power, reactive power, energy	15 min, 5 min, 1 min, 10 s	Missing intervals, noise, rounding	Load forecasting, clustering, customer profiling
Electrical parameters	Voltage, current, power factor, frequency	1 s to 5 min	Sensor noise, voltage fluctuations, calibration errors	Power quality monitoring, anomaly detection
Event driven data	Outages, sags, swells, alarms, restoration events	Event based	Timestamp accuracy, incomplete logging	Reliability assessment, grid disturbance analysis
Cross domain energy data	EV charging, PV output, ESS SoC, HVAC cycles	1 s to 1 min	Synchronization issues, heterogeneous formats	Flexibility modelling, integrated forecasting
Metadata and contextual data	Tariff type, building attributes, weather data	Static or hourly	Inconsistencies, incomplete metadata	Model enhancement, interpretability, segmentation

policy and infrastructure planning [11]. This kind of insight is particularly valuable as utilities seek to forecast and manage the increasingly complex interactions between customers and DERs.

In addition to clustering, feature selection plays a critical role in reducing model complexity and enhancing performance, particularly when working with high-dimensional datasets. Feature selection methods aim to identify the most relevant attributes in a dataset that contribute meaningfully to prediction or classification outcomes. This is especially important in machine learning applications, where including irrelevant or redundant features can lead to overfitting, a condition in which the model performs well on training data but poorly on unseen data. For example, Al Khafaf et al. [12] used a genetic algorithm to isolate the most predictive features from a dataset covering over 600 residential customers in Victoria, Australia, resulting in improved model accuracy and reduced computational time.

While feature selection has traditionally been regarded as an essential step for reducing computational cost, simplifying model structure, and mitigating overfitting risks, recent perspectives highlight that its widespread use often reflects the limitations of conventional machine learning models rather than an inherent need to eliminate features. High resolution smart meter datasets contain subtle temporal and cross-variable interactions that may not appear important in low-resolution or aggregated analyses but can significantly influence sub-hourly fluctuations, event signatures, or behavioural micro-patterns. Removing such variables may simplify the modelling process but can also conceal latent dependencies that are relevant for high fidelity tasks. As noted by Jiang et al. [13], deep Bayesian calibration frameworks demonstrate that high-dimensional parameter spaces can be managed effectively through representation learning instead of manual variable pruning. These approaches learn hierarchical embeddings that preserve multi-scale relationships, thereby reducing reliance on feature elimination and allowing the model to retain information that might initially appear insignificant. From this perspective, feature selection is often a compromise imposed by model architecture constraints rather than a universal best practice. Emerging deep learning and probabilistic modelling techniques shift the emphasis from selecting features to representing them efficiently, suggesting that future analytics pipelines may prioritise feature transformation and representation over explicit elimination, especially when working with fine-grained smart meter datasets.

Another important consideration in pre-processing is the temporal alignment and synchronization of datasets. Since smart meters, sensors, and other grid devices may operate on different sampling intervals or data transmission schedules, aligning their outputs into a coherent time series is essential. Improper alignment can obscure cause-effect relationships or lead to misleading correlations in models. Techniques such as interpolation, resampling, and time-window aggregation are commonly used to standardize data and enable meaningful cross-variable analysis.

As smart meter datasets continue to grow in scale and complexity, automated pre-processing pipelines are becoming increasingly valuable. These pipelines integrate various pre-processing tasks, from anomaly detection and missing data imputation to feature engineering and

normalization, into a streamlined workflow. They enable utilities and researchers to maintain consistent data quality across large-scale deployments and support the reproducibility of analytical results. By investing in robust and intelligent data pre-processing systems, DNOs can unlock the full potential of smart meter data, enabling more accurate forecasting, effective load management, and resilient grid operations. Fig. 2 illustrates the typical flow of electricity analytics, where smart meter data undergoes pre-processing, such as outlier removal, feature selection, and clustering, before being used in analytical models that support applications like demand forecasting, theft detection, and outage detection.

2.3. Data uncertainty and behavioral variability

Smart meter analytics increasingly depend on the premise that digital measurements faithfully mirror underlying consumption patterns, device interactions, and network conditions. In practice, however, the reliability of these measurements is shaped by uncertainty that emerges from technical, operational, and behavioral factors. These uncertainties influence the performance of forecasting models, demand management systems, anomaly detection frameworks, and grid-support applications. A rigorous understanding of uncertainty and behavioral variability is therefore essential for developing robust and trustworthy energy analytics.

A primary category of uncertainty stems from instrumentation characteristics. Smart meters differ significantly in hardware design, internal sampling processes, filtering strategies, and calibration procedures. These differences introduce measurement drift and signal distortion that can compromise data consistency. High frequency appliances, inverter-based resources, and devices with fast switching patterns generate waveform irregularities that may not be adequately captured by meters with limited dynamic response. Empirical work on smart meter evolution and measurement accuracy show that quantisation, sensing limits, and instantaneous noise can introduce discrepancies between actual load conditions and recorded values, with implications for the reliability of downstream metrics and diagnostics [14,15].

Communication and data transfer limitations create a second axis of uncertainty. Many smart metering systems depend on mesh networks or low bandwidth communication protocols that can experience congestion, interference, or intermittent connectivity. As a result, data gaps appear as isolated losses or extended intervals of missingness, often coinciding with adverse conditions or local network stress. Analyses of smart meter datasets show that missing data often arise from system-level issues such as network interruptions or power outages, rather than user behavior, which cautions against naïve interpretations that treat gaps as behavioral anomalies [16,17]. Reconstruction techniques such as probabilistic matrix factorisation, expectation maximisation, and transformer based imputation can mitigate these disruptions, although they introduce model driven uncertainty that must be quantified to maintain analytical integrity.

Human behavior introduces another major source of variability. Consumption patterns fluctuate with occupant routines, socioeconomic conditions, occupancy changes, and personal preferences. Irregular



Fig. 2. Flow of data pre-processing and applications in electricity analytics.

appliance usage, spontaneous electric vehicle charging, and unexpected thermostat adjustments produce short term fluctuations that challenge predictability. Studies using high resolution smart meter data show that households with similar structural characteristics may exhibit markedly different load signatures due to behavior driven randomness [18]. These dynamics complicate the development of generalizable models, particularly for applications that require fine grained temporal precision such as short term forecasting, flexibility estimation, and event detection.

Automated and semi automated systems add further complexity. Smart appliances, demand response controllers, and energy management systems often modify operating schedules in response to price signals, occupancy sensing, or comfort algorithms. Such autonomous adjustments disrupt historical patterns and introduce adaptive uncertainty. Studies on intelligent demand response and coordinated energy management frameworks indicate that automated control and multi-device coordination significantly alter load dynamics, leading to more stochastic and variable consumption patterns, particularly during peak-demand events and coordinated control actions [19].

Operational and environmental conditions within the distribution grid introduce additional uncertainty. DERs, such as photovoltaic systems, batteries, and electric vehicle chargers, create rapid fluctuations and transients that shape both consumption signals and grid behavior. Voltage fluctuations, harmonic distortion, transients, and other power-quality disturbances inherent to modern distribution networks introduce measurable deviations in smart-meter readings. Empirical comparisons with reference power-quality analyzers show that network-driven electrical phenomena can distort voltage, current, and power measurements, underscoring the need to account for power-quality effects when interpreting smart-meter data [20].

These diverse sources collectively influence the propagation of uncertainty throughout the analytical pipeline. Once uncertainty enters the measurement layer, it affects feature extraction, model training, and inference outcomes. Machine learning and deep learning models, especially those with non linear architectures, may amplify small perturbations. This poses risks for operational tasks such as automated control, distributed energy resource scheduling, and flexibility dispatch. In response, contemporary energy analytics increasingly incorporate uncertainty aware modelling approaches.

Modern mitigation techniques rely on probabilistic modelling, Bayesian inference, and ensemble learning. Probabilistic forecasts express outcomes as distributions rather than point estimates, which enables operators to reason about risk and confidence. Advances in Bayesian deep learning, including variational methods and Monte Carlo based approximations, have strengthened the ability to estimate both parameter uncertainty and data related uncertainty in load forecasting environments [21]. Ensemble learning approaches, particularly hybrid deep learning ensembles, help stabilize predictions under behavioral variability while reducing sensitivity to measurement noise [22]. A parallel development is the integration of multimodal data sources, for example occupancy sensors, meteorological streams, and appliance metadata. These auxiliary data channels provide contextual structure that constrains uncertainty and supports more robust inference [23,24].

Understanding and managing data uncertainty and behavioral variability is therefore fundamental for reliable smart meter analytics. By recognising the multiple sources of uncertainty and applying advanced mitigation strategies, DNOs can improve the resilience, accuracy, and trustworthiness of data driven decision systems within future energy infrastructures.

3. Data-driven applications for distribution network operation

This section presents a thematic review of key applications where data-driven methods are actively being developed and deployed. These include short- and long-term demand forecasting, electricity theft detection, outage detection and localization, anomaly detection, topology identification, and the integration of emerging technologies such as

EVs and energy storage systems (ESSs). Across each of these domains, the use of smart meter data, coupled with novel algorithmic approaches, is redefining how DNOs manage network behavior, respond to events, and plan future investments. The following subsections explore each application area in detail, highlighting the evolving methodologies, practical challenges, and research opportunities that define the modern landscape of distribution network operation. Fig. 3 illustrates the role of smart meter data as the central enabler for a range of analytics-driven applications in distribution network operations. It highlights how data collected at the grid edge feeds into critical functions such as demand forecasting, outage detection, topology identification, and EV integration, each supported by distinct analytical methods.

3.1. Demand forecasting

Electricity demand forecasting is pivotal in smart metering systems for ensuring effective power system management. It enhances energy monitoring, optimises energy usage for households and buildings, promotes electricity conservation, mitigates environmental impacts, and bridges the gap between supply and demand. Accurate demand forecasting also plays a vital role in facilitating stakeholder decision-making and strategic planning for the future [3,25–28]. Demand forecasting can be categorised based on the time horizon of prediction, such as ultra-short-term, short-term, medium-term, and long-term forecasting [29]. Alternatively, forecasts may focus on specific temporal scales, including hourly [30,31], daily [32,33], monthly [34], or annual intervals [35].

Various techniques have been employed for demand forecasting, including Artificial Neural Networks (ANNs) [25], Support Vector Regression (SVR) [36] and Regression Tree [30]. Additionally, hybrid models that integrate multiple techniques have been proposed to improve forecasting accuracy [37]. These models address specific forecasting challenges from different perspectives, although no single model is universally optimal across all scenarios. Since electricity consumption data is inherently time-series in nature, Recurrent Neural Networks (RNNs) [38] based deep learning approaches such as Long Short-Term Memory (LSTM) [34,39] and Gated Recurrent Unit (GRU) [37] have gained popularity for demand forecasting tasks. These models often deliver superior performance, especially for multi-step ahead predictions. Furthermore, models like Deep Belief Networks (DBNs) have also demonstrated effectiveness in this domain [33]. The performance of data-driven models is influenced by various factors, including unpredictable environmental uncertainties, significant variability in the operational conditions of system units, nonlinear degradation patterns, the availability of monitoring sensors, and the volume of data samples. A single model, even when optimally configured for a specific dataset, may not consistently yield satisfactory results. In such cases, ensemble learning, which combines the strengths of multiple models, can offer a robust alternative by enhancing generalisation and predictive accuracy compared to individual models [39–41].

To achieve precise demand forecasting, numerous variables can be considered based on the specific forecasting targets. These variables may include calendar-related information, such as weekdays, weekends, and hours of the day [42,43], electricity price [33], special events, such as FIFA World Cup, and economics [44], weather information [45] such as temperature, wind speed humidity. Key statistical measures, such as the maximum, minimum, median, and average values of demand and weather data, have been employed to derive meaningful features for predictive models [46]. However, not all variables are relevant in every scenario. Some variables may be redundant or noisy, adversely impacting prediction accuracy and computational efficiency. Consequently, recent studies have explored feature selection techniques [43, 47], such as cross-correlation and mutual entropy, to eliminate extraneous variables and improve model performance. In addition to feature selection, data transformation techniques can enhance forecasting accuracy by converting raw data into different domains. Methods such as

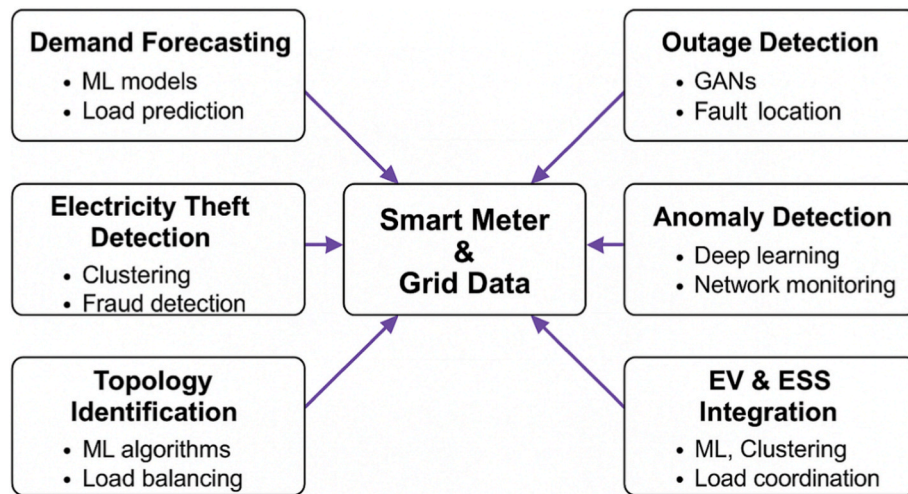


Fig. 3. Overview of data-driven applications in distribution network operations.

Principal Component Analysis (PCA) [42,46], which reduces dimensionality by transforming data into principal components, and Discrete Wavelet Transform (DWT) [48], which decomposes data into different resolutions, have been successfully applied to electricity demand forecasting. Recently, Evolutionary Algorithms (EAs) have shown promise across various aspects of demand forecasting due to their ability to solve complex combinatorial optimization problems. EAs have been employed for feature selection [42,43,45], ensemble learning [46], and parameter optimization [31]. Nevertheless, determining the best approach among these methods remains challenging without comprehensive comparisons under identical conditions, as each technique continues to evolve and refine its capabilities.

Recent advancements in demand forecasting methodologies emphasize the integration of hybrid models and intelligent optimization techniques to address the complexities of modern power systems. Dong et al. [49] proposed a short-term power load forecasting system combining data decomposition, deep learning architectures such as CNN and Bi-LSTM, and weighted linear error correction to handle the nonlinear and variable nature of load data. Their approach, tested on Australian datasets, demonstrates robust accuracy and adaptability to diverse grid conditions by combining point and residual forecasting values. Similarly, Al-Ja'afreh et al. [50] enhanced forecasting accuracy using a multi-stage framework that integrated CNN-LSTM models with advanced feature selection and hyperparameter tuning. This method reduces errors across multiple network levels, including residential, feeder, and substation scenarios, making it suitable for active network management and the integration of renewable energy sources. Complementing these studies, Zulfiqar et al. [51] presented a hybrid framework combining Locally Weighted Support Vector Regression (LWSVR) with adaptive grasshopper optimization (AGO) and novel feature engineering to refine input features and optimize model parameters. Their approach achieves superior stability, convergence, and accuracy in short-term load forecasting when applied to datasets from Australia and the U.S., outperforming traditional and hybrid models. Together, these studies highlight the critical role of hybrid models, intelligent optimization, and feature engineering in managing the challenges of nonlinear load patterns, renewable energy integration, and grid variability.

The integration of exogenous variables, including weather and behavioral indicators, further enhances forecasting accuracy. Studies show that incorporating temperature, humidity, solar radiation, and day-type indicators (e.g., weekday vs. weekend) results in improved performance across models such as ARIMAX, TBATS, and NARX [31, 44]. Peplinski et al. [52] found that building square footage and the month of the year were among the most influential predictors, revealing

seasonal and structural dependencies. Meanwhile, Kim et al. [53] used time-series clustering to group households with similar consumption behaviors, followed by the application of a suite of statistical and neural forecasting models. Their findings confirmed that cluster-based forecasting, especially when combined with neural models like NARX, outperformed general models by capturing distinct household usage patterns. This cluster-based approach reduces the variance in the data, thereby enabling more precise modeling of specific customer segments.

Hybrid deep learning models that merge multiple architectures have emerged as a dominant trend in recent literature. These models address the limitations of traditional forecasting by adapting to large volumes of high-dimensional data. For example, a recent study by Mbey et al. [54] introduced a hybrid model comprising Feed-Forward Neural Networks (FFNN), LSTM, and Multi-Objective Particle Swarm Optimization (MOPSO). Their model achieved high prediction accuracy ($R^2 = 0.999$, $RMSE = 1.15$) for both long-term demand forecasting and solar PV generation using smart meter and climate data collected in Douala, Cameroon. The use of multi-objective optimization enhanced the convergence and generalization of the deep learning framework, underscoring the growing relevance of such techniques for holistic smart grid planning.

Moreover, the operational application of demand forecasting extends beyond consumption prediction into grid optimization, tariff design, and distributed energy resource management. High-resolution forecasts enable utilities to design dynamic pricing strategies and identify load shifting opportunities for demand-side management. They also inform distributed energy resource scheduling, particularly for prosumers with rooftop solar or battery storage. For instance, the model by Mbey et al. [54] integrates PV generation forecasting alongside load forecasting, which allows utilities to balance supply and demand more effectively in real-time while accounting for intermittent generation.

Demand forecasting remains a cornerstone of modern distribution network operation, increasingly driven by advances in machine learning and data availability. The integration of smart meter data, exogenous variables, and hybrid deep learning frameworks has yielded significant improvements in accuracy and operational relevance. New approaches such as time-series clustering and optimization-enhanced neural networks are enabling more nuanced and localized forecasts. As distribution networks evolve toward greater decentralization and digitization, accurate, adaptive, and scalable demand forecasting models will be indispensable for ensuring grid stability, improving efficiency, and enabling the energy transition.

3.2. Electricity theft detection

Electricity theft in modern distribution networks poses a significant operational and financial risk to utilities worldwide. Theft detection in the electricity grid involves identifying unusual activities such as tampering with meter readings, bypassing electricity meters, or hacking to manipulate consumption measurements [55,56]. These practices are often driven by socio-economic conditions, including high energy costs, unemployment, and broader economic instability. Beyond the economic implications, theft introduces substantial public safety concerns, such as fires, electrocution, and equipment damage. It also disrupts the stability of the distribution network, leading to voltage fluctuations and fault propagation, which ultimately affect the quality and reliability of power delivery [57,58].

Conventional supervised learning models such as Support Vector Machines (SVMs) have been employed for theft detection. However, these models require access to detailed customer consumption data, increasing exposure to data breaches and exacerbating privacy concerns [59,60]. Additionally, they often underperform in dynamic or imbalanced environments where fraudulent behavior evolves over time. Traditional field inspections, though effective, are resource-intensive and infeasible for large-scale implementation, especially in urbanized areas with dense infrastructure [61]. Consequently, the focus has shifted to data-driven strategies that utilize smart meter data and machine learning to identify anomalies indicative of electricity theft, offering a more scalable and sustainable solution.

To improve detection efficiency, advanced inspection algorithms have been developed. Xia et al. [62] introduced the Suspicion Assessment-Based Inspection (SAI), which ranks users based on anomaly likelihood using both historical records and deviations from expected usage. This binary-tree-based prioritization reduces the cost and time required for inspections while maintaining high detection rates. Similarly, Massafiero et al. [63] proposed a priority ranking system based on user consumption profiles and utility-specific cost functions, streamlining enforcement actions by targeting the most suspicious cases first. These targeted approaches are particularly effective when resource allocation is constrained, allowing utilities to focus on the highest-risk areas without blanket surveillance.

Hybrid models that combine supervised and unsupervised learning offer further performance improvements in electricity theft detection. Zheng et al. [64] presented a model that integrates Maximum Information Coefficient (MIC) analysis with the Clustering by Fast Search and Find of Density Peaks (CFSFDP) algorithm. This approach detects irregular consumption behavior with high accuracy (96 %) by uncovering latent patterns among suspected consumers. In contrast, Gao et al. [65] proposed a technique based on active power and voltage magnitude fluctuations rather than direct consumption metrics. This method bypasses smart meter manipulation by leveraging grid-level electrical parameters to infer anomalies, offering resilience against meter-specific attacks.

Deep learning approaches are increasingly adopted for energy theft detection because of their capacity to model complex, high-dimensional consumption patterns. Neural architectures such as CNNs, RNNs, and LSTMs have demonstrated high accuracy in identifying anomalous usage behavior [66–68]. Traditional SVM-based techniques [69] continue to be relevant but often fall short when dealing with non-linear consumption trends. Meanwhile, game theory approaches have been applied to model strategic interactions between utilities and potential thieves [70], while statistical methods analyze load profiles and residuals to flag outliers [59,71,72]. The choice of method ultimately depends on the utility's infrastructure, available data granularity, and desired balance between computational complexity and detection accuracy.

Recent advancements also emphasize privacy-preserving detection frameworks that operate in real time. For example, the MOVSTAT-GAT model leverages spatio-temporal graph attention mechanisms to detect

false data injection attacks (FDIAs) at localized levels using unsupervised learning. It captures short-term and geographically constrained attacks more effectively than conventional LSTM-based models, significantly improving early detection performance [73]. In parallel, a decentralized federated learning approach that integrates threshold homomorphic encryption with XGBoost has been proposed to preserve consumer privacy while enhancing detection accuracy. This method allows distributed utilities to collaboratively train theft detection models without sharing raw consumption data, overcoming common challenges such as data imbalance and non-IID distributions [74].

Complementing these privacy-aware frameworks, recent studies have explored hybrid, optimization-enhanced architectures. For example, Mbey et al. [75] introduced a hybrid SVM-PSO algorithm trained on real smart meter data to detect abrupt changes in consumption. Their model achieved 98.9 % detection accuracy and incorporated calendar-based contextual features, such as holidays and seasonal patterns, to improve interpretability. Similarly, Xia et al. [76] presented the ETD-ConvLSTM model, which processes spatio-temporal sequences of electricity usage to detect anomalies with high precision. These models demonstrate that the integration of temporal awareness, optimization techniques, and contextual variables significantly enhances model performance in diverse operating environments.

Electricity theft detection has matured into a complex, multidisciplinary field that leverages smart meter data, machine learning, and privacy-preserving protocols. As smart grids continue to expand, detection strategies must evolve to address emerging threats, data privacy regulations, and scalability challenges. The trend toward federated, decentralized, and context-aware AI systems marks a significant shift in how utilities approach theft prevention. By combining real-time monitoring, anomaly detection, and targeted inspection methods, future systems will be better equipped to maintain grid security, operational efficiency, and consumer trust.

3.3. Outage detection

Effective outage management is important for distribution networks and has been significantly enhanced by the widespread adoption of smart meters [77]. A power outage in the smart grid refers to the interruption of electricity supply, often caused by issues such as damage to distribution lines [77–82], adverse weather conditions [83], equipment failure [84], or cyber-physical attacks [85], resulting in critical effects on the distribution networks. These outages have a direct impact on stakeholder satisfaction, particularly customers, making reliable outage detection essential for improving grid stability and reliability [86]. Advanced metering infrastructure (AMI) and data analytics offer the potential to enable near real-time detection of outages and improve system recovery time through rapid fault localization and service restoration.

Research efforts have explored various data-driven and model-based approaches to detect outages in distribution lines. IEEE test feeders [77, 78,81,87] are widely used to simulate and validate outage detection scenarios. One line of investigation involves dynamic programming-based sensor placement, which addresses the computational complexity of identifying simultaneous outages. These solutions integrate multiple sensor types and utilize polynomial execution time algorithms to detect arbitrary load variations under varied topological conditions [81]. Meanwhile, Bayesian probabilistic methods have demonstrated robust performance in detecting multiple simultaneous outages under random scenarios, including tests on IEEE-14 and IEEE-57 bus systems [88].

Real-time outage detection has also benefited from data fusion techniques. One approach validated using Pacific Northwest National Laboratory (PNNL) feeder data combined line sensor outputs with periodic smart meter readings to enhance visibility into outage events [79]. Predictive algorithms using phase measurement unit (PMU) data have also been explored to address challenges in detection timing and

sampling intervals [80]. These methods effectively increase the temporal resolution of detection by compensating for sampling limitations through data interpolation and forecasting. In parallel, compressive sensing techniques have been proposed to extract outage signals from sparse PMU data by leveraging voltage phase angle correlations [89].

Other innovative strategies include matrix analysis techniques that interpret the power network as a graph and detect line outages through structured changes in connectivity patterns [90]. These methods are particularly useful for identifying both single and multiple line failures and can operate effectively with limited real-time data. Additionally, rule-based methods have been used to mine outage patterns from historical data. For instance, Apriori-based association rule mining applied to outage, weather, and load data has shown promise in identifying equipment-related outages using chi-square and lift index scores [84]. These association-driven insights contribute to the development of intelligent alarm systems and targeted maintenance strategies.

The increase in high-frequency smart meter data has driven the development of streaming data frameworks designed for near-instant outage detection. Such frameworks support real-time analytics pipelines and improve response time for field service crews by automatically initiating alerts and fault localization procedures [91]. In tandem, cybersecurity researchers have studied the impact of local cyber-physical attacks that lead to line disconnections, proposing graph-based and AI-driven methods to localize the resulting outages within the grid [85]. These studies underscore the intersection between physical infrastructure vulnerabilities and cybersecurity risks, reinforcing the need for integrated anomaly and outage detection platforms.

Generative adversarial networks (GANs) have recently gained traction as powerful tools for outage detection in data-limited environments. In particular, GAN-based methods have been applied to partially observable distribution systems by learning the baseline data distribution and flagging significant deviations as potential outages [82]. Yuan et al. [92] proposed a GAN-driven framework that partitions the grid into zones and uses overlapping node pairs to estimate normal voltage distributions. Anomalies are detected as deviations from this learned baseline. This approach enables accurate outage detection even with incomplete smart meter deployment and improves scalability by removing the need for supervised labels. Similarly, Liao et al. [93] introduced the quick line outage identification (QLOI) method, which uses statistical changes in voltage dependencies for fast detection without requiring full network observability.

Despite these advancements, efficient outage localization remains a challenging task due to the heterogeneous nature of data, limited observability in legacy systems, and increasing network complexity. The integration of AI models, such as GANs and statistical change-point detectors, provides scalable and adaptive solutions. However, their performance depends heavily on the availability and quality of training data, which remains a key limitation. Ongoing research is needed to address these gaps, particularly through the development of semi-supervised and transfer learning techniques that can generalize across diverse network conditions and topologies.

3.4. Anomaly detection

Anomalies in power distribution systems arise due to a variety of issues, including hardware or software malfunctions, electrical abnormalities, power surges, adverse weather conditions, equipment failures, or security attacks. Such deviations from normal operation are collectively referred to as anomalies. Detecting these anomalies is crucial for ensuring the reliability of power distribution systems and delivering benefits to customers, network operators, and the power grid as a whole [94]. Anomalies not only degrade power quality but also compromise grid stability and can be indicative of deeper systemic issues, including cyber threats or latent equipment degradation. With the increasing deployment of smart meters and real-time monitoring infrastructure, anomaly detection is now a core component of intelligent distribution

network management systems.

Numerous studies have developed algorithms to detect and manage these anomalies. A state estimation-based method, for instance, was developed to identify meter tampering and quantify non-technical losses (NTL) like energy theft, thereby reducing losses and improving data trustworthiness [95]. This method uses IEEE 13-bus test feeder data and real-world energy consumption records to assess feeder-level losses accurately. In another approach, daily power supply and electricity sales data were analyzed to systematically identify anomalies in both NTL and technical losses (TL), focusing on the nature, time, and spatial location of these irregularities [96]. These methods highlight the importance of data granularity and system-wide visibility in improving anomaly detection accuracy.

Equipment-related anomalies, such as cable degradation or power line faults, have been a particular area of interest due to their direct impact on reliability and safety. A notable approach employed high-frequency signals, ranging from a few kHz to several MHz, using power line modems as sensors to detect and classify anomalies [97]. Two detection algorithms were developed to locate faults based on variations in signal characteristics. In another study, a spatio-temporal matrix for each feeder line was constructed, and variations in its covariance matrix spectrum were used to develop anomaly indicators [98]. These indicators were able to capture anomalies by analyzing changes in spatio-temporal correlations, showing strong performance across both synthetic and real-world datasets. In a similar context, a rule-based system was proposed to detect transformer connection errors and recommend switching strategies for balancing loads and mitigating transformer overloads [99].

Cyberattacks are a major concern in modern distribution networks, and anomaly detection plays a central role in identifying these threats. Attacks such as false data injection (FDI) can compromise decision-making, resulting in financial losses and operational disruptions [100]. To address these threats, dynamic analytical frameworks have been developed to assess intra-interval operational security risks caused by FDI attacks [101]. These models incorporate real-time system variability to offer more responsive protection. Another method utilized a joint-transformation approach based on measurement dynamics, combined with Kullback–Leibler divergence, to detect FDI anomalies by comparing probability distributions before and after potential attacks [101,102]. This method was validated using IEEE standard test feeders. A different study adopted a 24-step load forecasting model followed by multiphase optimal power flow (OPF) analysis to identify power flow abnormalities, further extending detection capabilities to complex system conditions [103]. A cyberattack estimation technique based on naive Bayes classification was also introduced to reconstruct attack benchmarks using neural network-based load forecasting, which was evaluated through Monte Carlo simulations [104].

Expanding beyond detection, operational risk assessment frameworks are now incorporating unsupervised learning models to classify the severity of detected anomalies. One recent approach utilizes bidirectional generative adversarial networks (BiGANs) to extract features from real-time monitoring data and classify operational states into emergency, high-risk, preventive, or normal categories [105]. These classifications are made using confidence intervals derived from historical operational baselines. Additionally, PMU data has been leveraged to detect grid events, providing enhanced situational awareness for operators through anomaly-based alerts [106,107]. These applications underscore the role of anomaly detection not just in identifying disruptions but also in enabling intelligent grid decision support systems.

Smart meter data has also been repurposed for anomaly detection in residential and behavioral contexts. For example, Hernández et al. [108] used disaggregated smart meter data to detect deviations in daily routines such as sleeping and eating. Their system applies CNNs, LSTMs, decision trees, and random forests to predict expected appliance usage and flag anomalies. Similarly, Zhou et al. [109] developed unsupervised models using dynamic time warping and Mahalanobis distance metrics

to detect periods of abnormal inactivity, such as might occur from sudden illness or medical emergencies, in elderly households. These approaches demonstrate that anomaly detection extends beyond grid reliability and into the realms of healthcare and human safety monitoring.

Recent models have also adopted hybrid and cyber-physical strategies. Alkuwari et al. [110] proposed an optimized ConvLSTM model for identifying FDI attacks in smart grid environments, demonstrating a 91.3 % detection accuracy across multi-class attack scenarios. This vertical hybrid model captures spatial and temporal patterns in consumption data, significantly outperforming traditional classifiers. Meanwhile, Farooq et al. [111] introduced a confidence ellipse-based anomaly detector using smart meter state estimations. Their technique improves detection reliability by comparing observed measurements against statistical baselines, offering real-time capabilities and robust performance during multi-node cyberattacks. The increasing sophistication of these models reflects the growing importance of anomaly detection in ensuring the cybersecurity and operational integrity of distribution networks.

Despite considerable progress, anomaly detection in distribution networks remains an evolving field. The reliance on simulated data due to limited access to real-world operational events poses a challenge to model generalization [112]. Furthermore, anomalies related to low-frequency equipment faults or intermittent sensor failures are still difficult to detect with current datasets and models. Future directions include the development of universally accepted benchmark datasets, improved anomaly taxonomies, and explainable AI techniques to enhance trust and transparency. As distribution networks continue to integrate renewable energy, EVs, and demand-side flexibility, robust and adaptive anomaly detection frameworks will be essential to maintaining resilience and operational excellence.

3.5. Low-voltage network topology identification

Low and medium voltage distribution networks typically operate in a radial structure, where substations serve as roots and customer loads as terminal nodes. Topology identification in such systems involves detecting the status of switches, identifying active branches, and determining how loads are connected throughout the network. This process is fundamental to fault detection, service restoration, load flow estimation, and grid reconfiguration. However, operator-induced changes in switch statuses and outdated records often create discrepancies between the actual and documented network topologies. Therefore, accurate and real-time topology awareness is essential for reliable analytics, including state estimation and feeder-level optimization.

Advances in sensing technologies, including smart meters, PMUs, and Frequency Monitoring Networks (FNET), have substantially improved the observability of grid conditions such as voltage magnitude and frequency. These technologies enable operators to assess system health and topology more effectively [113]. Cunha et al. [113] proposed a multi-linear regression model that uses smart meter measurements and line conductor attributes to estimate both network topology and parameters, achieving 90 % mapping accuracy and 80 % accuracy for line characteristics. Mokhtar et al. further developed a tree-based machine learning approach to predict missing cable information with 85 % accuracy, although they noted that deeper integration of smart meter data would enhance model performance [114].

Other techniques have focused on correcting topology records within Geographic Information Systems (GIS). Algorithms in Refs. [115,116] applied graph theory and coordinate data from network components to identify connectivity errors with up to 97.8 % accuracy. These methods do not require parameter estimation and are therefore computationally efficient. Additionally, studies in Refs. [117,118] leveraged smart meter voltage magnitude data for transformer-to-customer mapping, an essential step toward full topology identification. These mappings provide foundational knowledge needed for grid modeling, particularly in

dense urban systems with complex branching structures.

Recent contributions have proposed innovative data-driven methods to improve topology identification. Jiao et al. [119] introduced a clustering-based method that processes real-time voltage data using t-distributed stochastic neighbor embedding (t-SNE) and DBSCAN clustering. This approach is resilient to noisy and incomplete smart meter data and successfully identifies topologies involving both customer connection phases and transformer-customer relationships. Their field tests on networks in Guangdong Province demonstrated high practicality and robustness in real-world conditions.

Handling partial smart meter penetration remains a significant challenge. Pengwah et al. [120] addressed this by proposing a two-step topology estimation framework. Their model first estimates the topology among smart-metered customers and then incorporates interval meters, those with limited measurement capabilities, into the estimated network using impedance modeling. This approach allows utilities with mixed metering infrastructure to still benefit from advanced topology identification without requiring full smart meter rollout.

Flynn et al. [121] developed an improved recursive grouping algorithm that accounts for fluctuations in transformer voltage, an assumption often ignored in prior models. Their algorithm uses smart meter-derived voltage sensitivity coefficients to build topology graphs and detect faults. Tested on Australian distribution networks, the model showed significant accuracy improvements compared to conventional methods and successfully localized faults using real smart meter data. These results validate the importance of accurate transformer modeling in topology estimation.

Xu et al. [122] proposed a topology identification method specifically designed to address the “same-bus-different-feeder” problem, where voltage correlation methods often fail. Their probabilistic graphical model integrates voltage and power dependency relationships and employs belief propagation to infer switch states accurately. This method overcomes the limitations of standard voltage-correlation approaches and improves inference efficiency in large, interconnected networks.

Finally, García et al. [123] introduced a Bayesian inference-based phase identification model. This method estimates each customer's phase connection using consumption profile variations and updates beliefs with new data over time. It achieved over 97 % accuracy on real European low-voltage feeders and can operate with or without prior labeled data. Such probabilistic models are especially useful in distribution systems where manual records are incomplete or outdated.

Despite these advances, several challenges persist in achieving robust, scalable topology identification in real-world distribution systems. A major limitation is the reliance on synchronized and high-quality data, which is not always available due to smart meter desynchronization, data gaps, or poor measurement quality [119,120]. Moreover, methods that depend on full smart meter coverage are not applicable in many networks with mixed or legacy metering infrastructure [120]. Another persistent challenge is managing computational complexity, especially in large-scale networks where exhaustive search or high-dimensional clustering becomes intractable. Additionally, many existing methods assume radial structures or static load conditions, which limits their applicability in dynamically reconfigured or meshed grids [122]. Validation remains another barrier, as ground-truth topologies are often unavailable or inaccurate, making it difficult to benchmark new methods. Field validation via inspections, drones, or dedicated instrumentation further raises costs. Addressing these issues will require hybrid approaches that fuse machine learning, optimization, and domain knowledge, while also advancing explainability and robustness to noisy or partial data.

3.6. Electric vehicles and energy storage integration

The integration of ESSs and EVs into modern distribution networks represents both a transformative opportunity and a formidable

challenge. These technologies enable greater flexibility, peak load management, and improved grid resilience. However, traditional distribution infrastructure was not originally designed to support the dynamic, bi-directional energy flows that result from large-scale deployment of EVs and ESSs. Their intermittent charging and discharging behaviors can introduce voltage instability, unbalanced loads, and complications in protection coordination, thus requiring advanced analytics and operational strategies [124,125]. Moreover, while ESSs are increasingly recognized as assets for grid support, their widespread deployment is limited by insufficient regulatory frameworks and uncertainties surrounding economic incentives and market participation.

This rapid growth in global EV fleet places increasing emphasis on efficient EV charging, a critical concern for users and policymakers alike, especially as infrastructure such as charging stations (CS) remains limited. Furthermore, EV charging behaviour significantly affects distribution network reliability. As EV penetration into the power system grows, electricity consumption profiles shift, leading to challenges such as energy loss, overloads, load imbalances, peak demand, voltage and current instability, and difficulties in coordinating protection systems. Consequently, research on EV charging technologies [126–128], the impact of EV integration [129,130], and optimal placement and sizing of charging stations [131,132] has intensified, aiming to save energy, enhance grid operation, and support utility grid expansion and planning.

EV charging technologies have been extensively studied with a focus on scheduling and strategy. Charging scheduling aims to optimize when vehicles are charged to balance grid constraints and user objectives [126,128,129,133–138]. These schedules minimize the total time and cost associated with EV charging, including travel, waiting, and energy costs, while ensuring user satisfaction regarding final state-of-charge (SoC) [133]. Online scheduling algorithms, real-time pricing models, and predictive methods such as day-ahead station capacity estimation have emerged as valuable tools to support both grid operators and consumers [134,135,138]. These models rely heavily on high-frequency consumption and mobility data collected from smart meters, CSs, and fleet management systems.

Charging strategies, by contrast, analyze consumer behavior, load profiles, and cost models to devise system-wide coordination plans [127, 139–142]. These strategies often incorporate game theory, user incentives, and social behavior modeling. For example, EV users demonstrating socially responsible behavior may delay charging to reduce congestion and stabilize voltage [142]. Temporal and economic factors, such as calendar constraints [140] and cancellation charges in dynamic pricing environments [129], are also integrated into optimization objectives. Data from smart charging networks can be used to simulate these scenarios, enabling adaptive strategies that account for realistic variability in usage and cost structures.

The development of charging station infrastructure is a key enabler for EV expansion. The siting and sizing of CSs directly influence service efficiency, grid stress, and user satisfaction. A novel graph-based model incorporating automorphic grouping has been proposed to synthesize EV networks and CSs under traffic constraints, maintaining waiting times below defined thresholds [131]. To optimize infrastructure placement, mixed-integer linear programming (MILP) models have been used to account for power system constraints, traffic density, and power line conditioning capabilities [132]. In parallel, predictive models that estimate the required number of CSs based on demand patterns and social cost minimisation have been introduced to guide long-term planning [143]. These approaches are increasingly supported by real-time traffic and charging behavior data, reflecting a convergence of energy and transportation analytics.

The joint integration of EVs and ESSs with distributed renewable energy sources presents further complexity. While these systems can support peak shaving, frequency regulation, and energy arbitrage, their stochastic behavior introduces new operational uncertainties. Effective integration will depend on coordinated infrastructure investment, real-time data acquisition, and cross-sectoral planning. Despite

technological progress, key challenges remain: lack of standardization in control interfaces, interoperability across different network elements, absence of real-time data visibility, and policy gaps related to energy storage compensation and grid service participation [125,129].

As EV and ESS adoption continues to accelerate, their integration into distribution systems will require collaborative efforts between utilities, regulators, and technology developers to ensure scalability, resilience, and consumer affordability. Addressing these multifaceted challenges will be vital for transitioning to a sustainable, data-driven smart grid environment. Future research should prioritise intelligent coordination mechanisms, such as vehicle-to-grid (V2G) communication, adaptive charging control, and co-optimization frameworks for ESSs and EVs. These developments must be informed by robust demand forecasting, load prediction, and network impact simulations.

3.7. Demand-side data analytics

Smart meters act as the primary data interface between end-users and the distribution network, providing continuous, fine-grained measurements that capture both technical consumption characteristics and behavioral patterns. While supply-side analytics mainly address generation, storage, and grid optimization, demand-side intelligence complements these efforts by revealing how users and buildings consume, adapt, and respond to energy dynamics in real time [144,145].

Building-level smart-meter data, typically collected at intervals ranging from 15 min to a few seconds, exhibit strong temporal and spatial variability driven by occupant behavior, equipment operation, and environmental influences. A growing research focus has been on non-intrusive load monitoring (NILM), where aggregated power signals are decomposed into appliance-level profiles. Early NILM studies used statistical pattern recognition [146], while more recent approaches employ deep-learning architectures such as convolutional and recurrent neural networks to improve accuracy and robustness [147,148]. These methods not only identify individual loads but also infer behavioral routines and enable real-time demand response participation.

Beyond load disaggregation, occupant behavior modeling has become essential for understanding energy variability and user flexibility. Studies have shown that clustering and probabilistic modeling techniques can effectively categorize consumption behaviors and identify response potential to price or comfort signals [149,150]. Such behavior-informed analytics support data-driven demand response programs, where aggregated flexibility from households and buildings can be predicted and coordinated with supply-side resources such as energy storage or distributed photovoltaic systems [151].

The integration of demand-side analytics with supply-side forecasting establishes a closed-loop information exchange between user behavior and grid operation. As shown conceptually in Fig. 4, smart-meter data enable bi-directional learning, where demand-side insights improve load forecasting accuracy, and supply-side feedback influences household-level optimization. Accurate modeling of building demand variability is widely recognized as a key enabler for short-term grid balancing and for improving the scheduling efficiency of storage and distributed generation systems [152].

Emerging research also emphasizes privacy-preserving analytics, particularly through federated learning frameworks that allow distributed model training without centralizing user data [153,154]. These techniques are crucial for scaling demand-side intelligence across large populations of prosumers while maintaining data confidentiality and regulatory compliance. Overall, incorporating demand-side analytics transforms smart-meter systems from passive monitoring tools into active decision-support infrastructures. By capturing the interplay among occupants, devices, and energy networks, these analytics provide actionable insights that support flexible, user-centric, and net-zero-ready energy systems [155].

Fig. 4 presents a conceptual overview of how demand-side and supply-side analytical processes interact through smart meter data. The

Demand-supply data interaction in smart-meter analytics

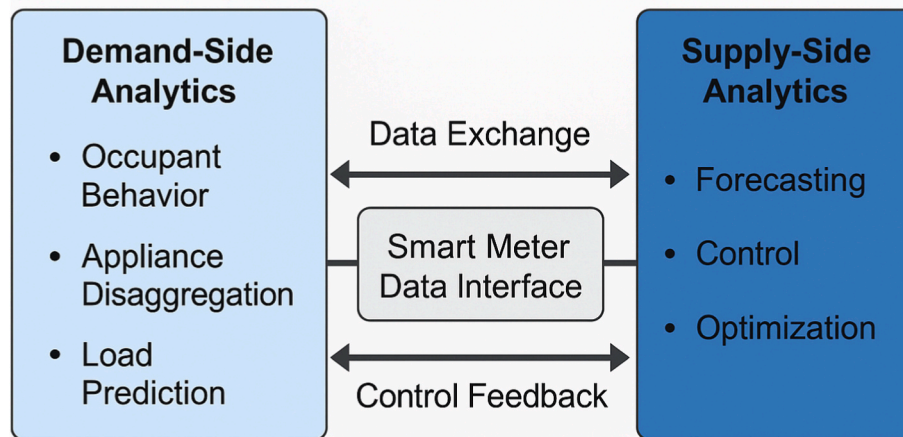


Fig. 4. Demand-supply data interaction in smart-meter analytics.

left panel groups key demand-side functions, including occupant behavior analysis, appliance disaggregation, and short-term or long-term load prediction. These processes generate detailed information about consumption patterns and user flexibility. The right panel shows supply-side analytical tasks, including energy forecasting, operational control, and grid or resource optimization. At the center of the diagram is the smart meter data interface, which serves as the point of connection for both analytical domains. Two directional arrows labeled Data Exchange and Control Feedback illustrate the flow of information. Data Exchange represents the transfer of consumption data from the demand side to supply-side functions. Control Feedback represents pricing signals, demand response instructions, or optimization adjustments that return from the supply side to influence user behavior or device-level operation on the demand side. The figure demonstrates how smart meter data establish a continuous feedback loop that supports coordinated, data-driven management of both consumption and system operations.

3.8. Data decomposition and aggregation for multi-scale smart meter analytics

Smart meter data contain overlapping signatures from appliances, occupant activities, and equipment operating cycles, which creates a complex composite load pattern that must be disentangled before meaningful analytics can be performed. Decomposition techniques address this challenge by separating aggregated electrical demand into constituent components that represent distinct behavioural and operational processes. The purpose of this separation is to expose patterns that would otherwise remain buried in the aggregate signal. For example, isolating HVAC consumption, lighting demand, and plug load fluctuations provides analysts with clearer insight into how occupants behave, how equipment operates, and how different loads contribute to peak consumption periods. Such decomposition enables a richer understanding of household energy dynamics and supports a range of applications that include behaviour modelling, equipment diagnostics, demand response targeting, and appliance-level efficiency assessment [156].

A central method used for decomposition is NILM. NILM analyses aggregate consumption and identifies the electrical signatures that correspond to specific appliances. Early NILM methods relied on detecting steady-state transitions, edge events, or simple power changes. Modern NILM frameworks leverage deep learning architectures that can recognise complex temporal relationships, overlapping appliance events, and non-linear dependencies within high resolution load traces

[157]. Convolutional networks, transformer layers, and hybrid models that combine signal decomposition with machine learning have significantly improved appliance identification accuracy. These approaches can capture subtle waveform characteristics and operate effectively even when user behaviour introduces irregularities. Research increasingly integrates NILM with probabilistic estimation, which allows models to represent uncertainty in appliance activation events and reduce misclassification in noisy conditions [158].

Decomposition extends beyond appliance identification. A growing body of work separates behavioural loads from operational loads. Behavioural loads arise from occupant decisions, such as cooking, entertainment, or the use of portable appliances. These loads show strong variability because they depend on personal preferences and daily routines. Operational loads, in contrast, originate from systems governed by physical or environmental conditions, including HVAC cycles, heat pumps, ventilation systems, water heaters, or electric vehicle charging. Separating these load types reveals deeper insights into the interplay between behaviour and equipment. Behavioural decomposition helps quantify stochasticity in consumption, while operational decomposition allows analysts to diagnose equipment performance problems or inefficiencies that are not easily visible in aggregate signals [159].

Aggregation operates as a complementary process to decomposition, transforming granular data into structured representations that match the temporal or spatial requirements of analytical tasks. Smart meters often generate sub-minute or minute-level measurements, which are essential for high-resolution modelling but can hinder long-horizon tasks due to noise and behavioural variability. Temporal aggregation addresses this by summarising data at hourly, daily, or weekly intervals. Hourly aggregation is widely used for forecasting and capacity planning, while daily and weekly aggregations reveal seasonal effects, long-term behavioural trends, and underlying energy routines. Aggregation does not merely reduce data volume. It enhances model stability by smoothing irregularities and aligning the dataset with the natural rhythms of system operation or policy cycles [160].

Spatial aggregation escalates this idea across households, feeders, neighbourhoods, or entire districts. When data from multiple households are aggregated, individual consumption peculiarities tend to cancel out, revealing dominant community-level patterns. District-scale aggregation supports urban energy planning, renewable integration studies, and demand response orchestration. Feeder-level aggregation provides distribution system operators with insights into transformer loading, voltage behaviour, and local congestion challenges, particularly in regions experiencing rapid uptake of EVs or rooftop photovoltaic

systems. Spatial aggregation thereby links household-level dynamics to broader operational phenomena, creating a coherent framework for grid-aware analytics [161].

A natural progression arising from both decomposition and aggregation is hierarchical modelling. These models connect multiple levels of analysis by linking appliance-level behaviour to household-level consumption, and household-level behaviour to feeder or district-scale dynamics. Hierarchical structures help researchers understand how small changes, such as adjustments in thermostat settings or appliance scheduling, propagate upward and influence system-wide patterns. Conversely, they show how system phenomena, such as feeder congestion or net-load variability from DERs, shape household-level behaviour or device operation. Multi-resolution modelling frameworks designed around these principles are increasingly used for forecasting, flexibility assessment, and integrated optimization of DERs [162].

Decomposition and aggregation provide a multi-layered lens for interpreting smart meter datasets. Decomposition uncovers fine-grained behavioural and operational insights, while aggregation contextualises these insights within broader temporal and spatial structures. Combined, they form a foundational toolkit for advanced smart meter analytics and support the growing need for scalable, interpretable, and system-integrated energy intelligence. A consolidated benchmark of the primary smart meter analytics methods discussed in Sections 3.1 to 3.5 is presented in Table 2 to provide a unified comparison across algorithms, datasets, metrics, and methodological characteristics.

3.9. Comparative framework of data-driven techniques

A fundamental insight emerging from the literature is that the

suitability of analytical techniques for smart-meter data intelligence depends more on the characteristics of the underlying data, such as temporal resolution, fidelity, completeness, and dimensionality, than on the intrinsic complexity of the model itself. Many previous studies have focused on how a method performs, but fewer have examined why certain algorithms are chosen under particular data conditions. This section therefore introduces a comparative framework that links data properties to algorithm selection and performance behavior.

Low-resolution datasets, for example hourly or daily readings, tend to smooth temporal fluctuations and measurement noise. For such data, conventional statistical models such as linear regression, ARIMA, and exponential smoothing remain robust, transparent, and computationally efficient. When the data frequency increases to the sub-hourly level, nonlinearities become more pronounced, and machine-learning models such as SVR, random forests, and gradient boosting demonstrate higher adaptability to diverse consumption patterns.

At high-resolution sampling below 1 min, temporal dependencies and stochastic behavior dominate. In such cases, deep-learning architectures such as RNNs, LSTM, CNNs, and hybrid combinations such as CNN-LSTM outperform classical techniques by capturing complex temporal and spatial relationships. However, these approaches demand extensive training data and computational resources, as well as rigorous data preprocessing to mitigate noise and missing intervals.

In addition, the completeness and reliability of data critically influence model choice. Techniques such as KNN, ensemble learning, or probabilistic inference are often preferred when dealing with noisy or partially missing datasets because of their tolerance to uncertainty. In multi-domain contexts where smart-meter data interact with distributed resources such as EVs, photovoltaics, or storage, hybrid or multi-agent

Table 2

Comparative summary of smart meter data analytics methods across forecasting, anomaly detection, theft detection, disaggregation, and behaviour modelling.

Application Domain	Algorithm Category	Typical Datasets	Common Metrics	Strengths	Limitations	Maturity Level
Short-Term Load Forecasting	Statistical models (ARIMA, SARIMA)	Hourly or 15-min data, multi-year	MAE, RMSE, MAPE	Simple, interpretable, low compute requirements	Poor performance on nonlinear patterns and DER-rich feeders	High (widely deployed)
	Machine learning (RF, SVR, GBM)	Household and feeder-level datasets	RMSE, R^2	Captures nonlinear and weather-driven effects	Requires feature engineering, sensitive to noise	High
	Deep learning (LSTM, GRU, TCN)	High-resolution smart meter data	RMSE, MAE	Learns temporal dependencies automatically, excels on dense data	Opaque models and high training cost	Medium–High
Anomaly and Event Detection	Rule-based and statistical thresholds ML classifiers (SVM, RF)	Monthly bills or hourly data Minute-level consumption traces	Detection rate, false-positive rate Precision, Recall, F1	Transparent and easy to deploy Effective for multi-pattern anomalies	Ineffective for complex anomalies Requires labeled datasets, feature selection overhead	High Medium
	Deep anomaly detectors (Autoencoders, GANs)	Sub-minute or high-variance datasets	F1-score, AUC	Handles subtle temporal distortions and unknown anomalies	Hard to interpret, prone to overfitting	Emerging
Energy Theft Detection	Statistical profiling	Customer billing + monthly meter data	Accuracy, Recall	Simple implementation	Poor for modern AMI with noise and variability	Medium
	ML models (RF, XGBoost)	AMI data with behaviour signatures	F1, ROC-AUC	High detection sensitivity	Vulnerable to class imbalance	Medium–High
	Deep learning (CNN, LSTM)	High-resolution consumption curves	F1, Precision–Recall	Detects temporal deviations and disguised theft	Needs large datasets with labeled theft	Emerging
Load Disaggregation (NILM)	Event-based NILM	High-frequency (kHz) datasets	F1, accuracy per appliance	Good for on–off loads	Inaccurate for overlapping or multi-state devices	Medium
	Deep NILM (Seq2Seq, CNN, Transformer)	1-sec to 1-min AMI data	MAE, SAE	Strong performance without feature engineering	High compute cost, privacy concerns	Emerging–High
Behaviour and Occupancy Modelling	Clustering (k-means, Gaussian mixtures)	Daily or hourly profiles	Silhouette score	Fast pattern discovery	Limited temporal context	High
	Sequence models (HMM, LSTM)	Minute-level smart meter data	Accuracy, log-likelihood	Captures routines and occupancy transitions	Requires sizeable datasets	Medium
	Probabilistic models (Bayesian, hierarchical)	Multi-scale datasets	RMSE, posterior uncertainty	Represents uncertainty and behavioural variability	Complex to tune	Emerging

Table 3
Data characteristics and suitability of analytical techniques for smart-meter applications.

Data Characteristic	Typical Analytical Methods	Representative Applications	Why It Fits	Key Limitations
Low-resolution (hourly to daily)	Linear regression, ARIMA, exponential smoothing	Long-term demand forecasting, energy planning	Captures coarse temporal trends and requires limited data	Misses short-term variability and has limited adaptability
Medium-resolution (15–30 min)	Support-vector regression, random forest, gradient boosting	Anomaly detection, short-term load prediction	Handles nonlinearities and balances data volume with interpretability	Requires feature engineering and may overfit small datasets
High-resolution (below 1 min)	LSTM, CNN, hybrid deep learning models	Event detection, high-frequency forecasting, appliance disaggregation	Learns complex temporal dependencies and dynamic behaviors	Computationally intensive and sensitive to noise or missing data
Incomplete or noisy datasets	KNN, ensemble models, Bayesian inference	Data cleaning, gap-filling, uncertainty estimation	Robust to outliers and capable of imputing missing intervals	Accuracy decreases when noise dominates and memory demand is high
Multi-domain or heterogeneous data (EV, PV, and building)	Hybrid or multi-agent learning, transfer learning	Integrated grid analytics, flexibility optimization	Captures inter-domain correlations and operational coupling	Requires large labeled datasets and raises privacy or synchronization challenges

learning frameworks are most suitable because they can model heterogeneous data flows and interdependencies across domains.

Table 3 summarizes how data characteristics shape analytical method selection and highlights the rationale behind each pairing. This framework moves the discussion from a *method-centric* to a *data-centric* perspective, clarifying why particular analytical strategies are effective within specific operational environments.

3.10. Cross-domain synergies and Trade-offs

Smart meter intelligence increasingly operates across interacting analytical domains rather than within isolated tasks. Forecasting, anomaly detection, outage inference, flexibility estimation, EV charging analytics, and ESS management share common data structures, temporal dependencies, and operational objectives. Cross-domain integration improves the robustness and interpretability of individual models and supports a shift toward system-level intelligence in distribution grids. Recent studies emphasize that such interactions are central to reliable and adaptive grid operations, particularly under high DER penetration and behaviour-driven variability [155,163].

Short-term load forecasting provides a natural foundation for multiple downstream tasks. Forecast residuals and uncertainty bounds have been shown to improve anomaly and outage detection by establishing expected load envelopes against which deviations can be evaluated [164,165]. Probabilistic forecasting further strengthens outage detection by enabling confidence-based localization of abnormal events in feeders lacking dense sensor infrastructure [166]. These synergies allow forecasting models to function simultaneously as predictive tools and as baseline detectors of cyber-physical disturbances.

EV charging and behind-the-meter ESS introduce distinct temporal signatures that interact with both forecasting and anomaly detection. Incorporating EV load patterns has been demonstrated to support improved peak-period forecasting accuracy and yield more stable prediction behavior [167], while ESS state-of-charge estimation enhances the detection of irregular charging or discharging behaviours that may indicate tampering, faults, or cyber intrusions [168]. Cross-domain EV–ESS analytics therefore help distinguish legitimate variability from suspicious activity in households and community-scale assets.

Demand response intelligence further links disaggregation, behavioural modelling, and flexibility estimation. Appliance-level load profiles derived from NILM improve the identification of schedulable loads and household flexibility potential [169]. Behaviour-aware models informed by smart meter data and comfort indicators improve the representation of user participation and comfort constraints, which are essential for designing responsive and equitable DR programs [170]. These flexibility insights carry operational value for outage handling and resilience, because flexible loads can be curtailed or shifted to assist grid recovery during faults.

Trade-offs emerge when multi-domain objectives compete for the

same data representations. Forecasting models often smooth consumption trajectories to reduce prediction error, yet this smoothing can remove higher-order temporal features needed for accurate anomaly detection or NILM [171]. Deep disaggregation models may prioritise appliance fidelity but overlook feeder-level features necessary for outage inference. Likewise, optimization algorithms for EV charging may flatten daily demand curves in ways that obscure subtle anomalies or distort behavioural inference. Multitask learning frameworks and shared latent-representation models offer promising pathways to balance these competing objectives by preserving both coarse and fine-grain signal characteristics [172].

As distribution networks evolve into data-rich, behaviour-coupled, and DER-dense environments, cross-domain synergies become essential. Smart meter analytics can be organized into coordinated pipelines in which load forecasting informs anomaly detection and validated anomaly signals support downstream scheduling decisions, such as EV and energy storage management. Integrated analytical frameworks of this kind enhance situational awareness, resilience, and operational efficiency by ensuring that outputs from one domain serve as actionable intelligence for another [167]. To clarify how analytical domains

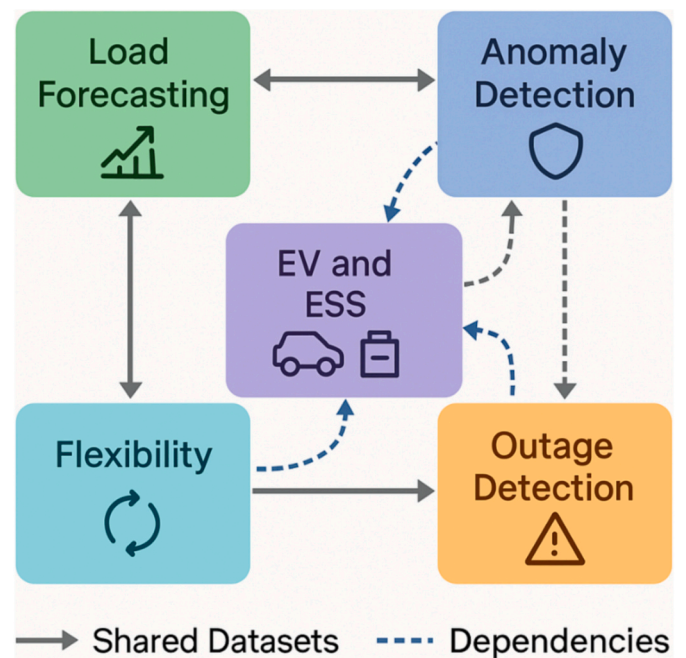


Fig. 5. Cross-domain interactions among key smart meter analytics tasks, showing shared datasets (solid arrows) and analytical dependencies (dashed arrows) across forecasting, anomaly detection, outage detection, flexibility estimation, and EV/ESS optimization.

interact within intelligent distribution networks, Fig. 5 synthesises the shared data flows and operational dependencies across forecasting, anomaly detection, outage detection, flexibility assessment, and EV/ESS analytics. The diagram highlights how outputs from one domain often become inputs or contextual signals for others, reinforcing the need for integrated, cross-domain smart meter intelligence.

4. Advancing network intelligence in future distribution systems

As operational complexity increases, with the integration of EVs, ESS, DERs, and flexible demand, traditional tools and architectures are proving insufficient. Network intelligence, driven by advanced analytics and real-time data processing, offers a path forward. It enables DNOs to transition from reactive operation to predictive and adaptive management, providing the foundation for greater reliability, resilience, and efficiency in increasingly dynamic environments.

However, the shift toward intelligent distribution networks is not without challenges. High data volume, inconsistent data quality, and privacy concerns create technical barriers, while interoperability gaps and the lack of regulatory clarity complicate deployment. Furthermore, as DNOs adopt technologies such as digital twins and fog computing to support localized, real-time decision-making, they must also invest in computational infrastructure and scalable analytics models. Balancing centralized and edge intelligence, ensuring trust in automated systems, and aligning operational goals with decarbonisation targets are critical to realizing the full potential of network intelligence. These developments mark a pivotal moment in the transformation of power distribution systems, where data becomes not just an asset, but the engine of future grid operation. Fig. 6 summarizes the key enablers and challenges associated with implementing network intelligence in modern distribution networks, illustrating how emerging technologies aim to address issues related to data quality, scalability, and operational complexity.

4.1. Issues in network intelligence

As distribution networks become increasingly digitized, the role of data-driven intelligence in grid operation has expanded dramatically. However, this evolution is not without its challenges. Section 3 highlighted the potential of advanced analytics across various domains,

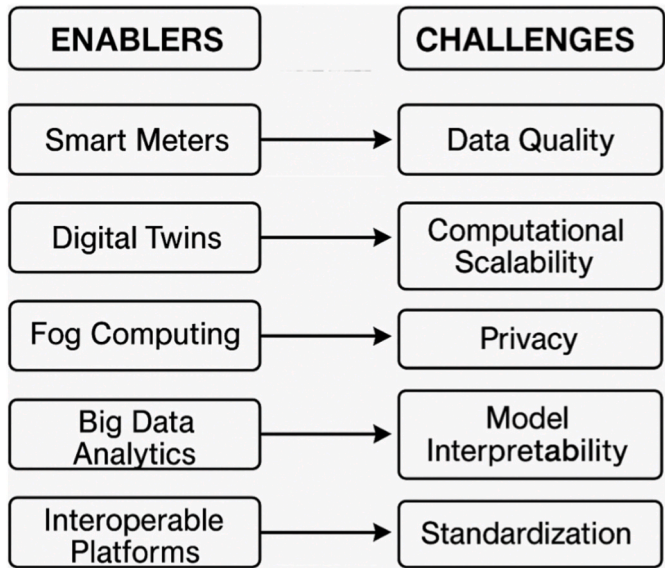


Fig. 6. Key enablers and challenges of network intelligence in distribution systems.

demand forecasting, theft detection, outage identification, anomaly detection, topology inference, and EV/ESS integration. Yet, each of these applications exposes critical gaps in the current state of network intelligence that must be addressed to ensure reliability, scalability, and resilience in future distribution systems.

A primary issue is data quality and availability. Many advanced algorithms require high-resolution, synchronized, and clean data to function effectively. In practice, utilities often deal with incomplete AMI rollouts, inconsistent data formats, latency in reporting, and privacy limitations. These constraints compromise the performance of machine learning models, particularly those reliant on deep learning or temporal analysis. Furthermore, data imbalance, such as the underrepresentation of theft or fault events, hampers the training of accurate and generalizable models.

Another concern is the interpretability and trustworthiness of AI models used in grid operations. While methods like neural networks and GANs offer impressive accuracy, their "black-box" nature raises operational concerns, especially for tasks like outage detection, where explainability is essential for safety and compliance. The lack of transparent decision-making hinders model adoption among engineers and regulators, particularly when outcomes affect grid reliability or customer billing.

Scalability and computational complexity also remain problematic. Algorithms that perform well in controlled test environments may fail to scale when deployed across large, heterogeneous networks. This is particularly evident in topology identification, where exhaustive search methods or dense clustering algorithms struggle with real-time implementation. Additionally, federated learning and distributed optimization, though promising, introduce coordination and synchronization challenges across utility infrastructures with varied capabilities.

The integration of emerging technologies, such as EVs and ESSs, introduces new dynamics that existing network intelligence systems are not fully prepared to handle. These assets are mobile, stochastic, and capable of injecting power into the grid, violating assumptions embedded in traditional models. Without adaptive learning mechanisms and coordinated control strategies, their operation may introduce volatility, undermine protection schemes, or worsen peak demand conditions.

Lastly, validation and ground truthing continue to limit progress. Many models are trained and tested on simulated or historical datasets with uncertain accuracy or outdated topology records. Field validation is costly and often impractical, especially for rare events like cyberattacks or simultaneous line outages. This limits confidence in model generalizability and hampers widespread deployment of advanced analytics in real-world settings.

Addressing these issues requires a multi-pronged approach, combining robust data governance, interpretable AI models, scalable architectures, and integrated planning frameworks. Only by overcoming these challenges can network intelligence systems mature into reliable tools for managing the complexity and uncertainty of modern distribution grids.

4.2. Network intelligence and net-zero carbon emission

Smart meter analytics play a measurable role in accelerating decarbonisation across distribution systems by improving energy efficiency, enabling flexible demand, reducing operational losses, and supporting the integration of low-carbon technologies. While earlier studies have illustrated these benefits qualitatively, recent research provides quantifiable evidence of efficiency gains, loss reductions, and demand flexibility enabled by advanced metering infrastructures and data-driven optimisation, which collectively translates to carbon reductions. These contributions arise through several pathways including enhanced load forecasting, abnormal consumption detection, targeted demand response programs, EV charging coordination, and operational loss minimisation [173,174].

Improved short-term load forecasting enabled by smart meter data directly reduces carbon emissions indirectly by increasing the accuracy of generation scheduling and reducing reliance on high-emission peaking plants and reserve resources. System studies indicate that improving forecasting accuracy by 10–20% can reduce reserve procurement costs by approximately 10%, with associated system-level emissions reductions translated in the low single-digit percentage range, depending on grid composition [175,176]. In systems with significant fossil-based ramping generation, scenario-based studies indicate that emissions impacts during peak demand and low renewable generation events may approach the upper end of translated reductions (on the order of several percentage points, and in some cases up to 10%), depending on system composition and assumptions [177].

Demand response (DR) programs built on smart meter insights contribute additional carbon savings by enabling more efficient demand management and strategic load adjustment. Recent evaluations indicate that household DR enabled by smart meter analytics can reduce electricity consumption by approximately 5–15%, with translated CO₂ reductions of 3–9% depending on the generation mix [178]. In markets where peak demand is served predominantly by fossil-based units, demand reductions can deliver a translated annualised carbon savings of up to 150–300 kg CO₂ per participating household [179]. Furthermore, behaviour-influenced DR programs guided by smart meter feedback can reduce household energy use by 4–7%, reinforcing long-term emissions reduction [180].

Energy theft detection and anomaly identification also contribute to decarbonisation by reducing NTLs. NTLs account for 1–15% of distributed electricity globally, depending on region. Smart meter analytics using machine learning have demonstrated 30–60% improvements in identifying theft cases, translating to avoided losses of 20–40 kWh per connection per month in high-risk networks [181]. These avoided losses translate to carbon savings of approximately 12–28 kg CO₂ per household per month under typical emission factors [182]. Although often overlooked in net-zero planning, reducing NTLs represents a significant pathway for emissions mitigation, especially in developing energy markets.

Another major contributor is the integration of EVs and behind-the-meter ESS, whose optimal operation depends on high-resolution meter data. Smart-charging strategies informed by meter data enable charging to be shifted toward periods of lower grid carbon intensity or renewable surplus, reducing charging-related emissions by approximately 20–40% in representative studies [183]. Fleet-scale analyses have demonstrated reductions of up to 0.1–2.5 tonnes of CO₂ per vehicle annually when smart charging is deployed consistently, depending on vehicle usage patterns and grid composition [184]. Similarly, ESS optimization based on smart meter analytics enhances self-consumption of rooftop solar, reducing grid import emissions by 18.8–20.8% during residential evening peaks in modeled scenarios [185].

Enhanced outage detection and improved distribution network reliability also contribute to carbon savings. By accelerating outage localization and restoration, smart meter analytics reduce backup generator usage and prevent unnecessary dispatch of fossil-fuel units. Field studies indicate that using smart meter data for outage analytics can reduce restoration times by 20–60%, depending on outage type, baseline response practices, and the degree of automation and smart meter integration, translating to avoided emissions of 5–12 kg CO₂ per customer per event depending on generator reliance [186].

When these pathways are combined, the cumulative decarbonisation potential becomes significant. Recent system-level modelling suggests that mature smart meter analytics ecosystems can reduce annual distribution-level emissions by 30% depending on DER penetration, market structure, and load flexibility [187]. These findings highlight that smart meter intelligence is not merely an operational enhancement but a strategic decarbonisation tool. Its quantitative contributions strengthen the argument for prioritizing advanced metering and data-driven optimization as integral components of net-zero transition

strategies. The quantitative ranges reported in Table 4 synthesize findings from multiple system studies and policy analyses; where direct emissions data are unavailable, energy impacts are translated into CO₂-equivalent values using standard emission factors. Resulting CO₂ ranges are indicative and depend on grid composition, operational assumptions, and regional emission factors.

Table 5 presents a summary of how key data-driven analytics domains contribute to net-zero objectives. It links each operational application, such as forecasting, outage detection, and EV integration, to its role in enabling renewable integration, reducing emissions, and improving grid efficiency, thereby highlighting the strategic value of network intelligence in decarbonizing power distribution systems.

4.3. Network intelligence and decarbonisation of the transport sector

The decarbonisation of the transport sector is inextricably linked to the successful integration of EVs into the power grid. As EV adoption accelerates globally, distribution networks are facing an influx of new, mobile, and often unpredictable demand sources. This shift represents a major departure from traditional, centrally managed grid operations and introduces both technical and operational complexities that current infrastructure is not fully equipped to handle. As examined in Section 3.6, network intelligence, powered by big data analytics, is essential for managing this transformation, yet several challenges remain.

One of the most pressing issues is the non-uniform and time-sensitive nature of EV charging behavior. Uncoordinated charging, particularly during evening peak hours, can lead to local transformer overloads, phase imbalances, and voltage instability. While charging scheduling and strategy optimization methods have shown promise (Section 3.6), their effectiveness depends on accurate real-time data, predictive modeling of user behavior, and dynamic coordination across thousands of endpoints. The lack of granular, high-resolution data in many regions, along with fragmented data ownership between utilities and private EV charging providers, limits the scalability of current solutions.

Another challenge lies in infrastructure planning and spatial coordination. The siting and sizing of EV charging stations must consider both traffic patterns and local grid constraints. As discussed in Section 3.6, data-driven methods, such as graph-based optimization and mixed-integer programming, have been developed to address these concerns.

Table 4
Quantitative carbon reduction potential enabled by smart meter analytics.

Smart Meter Analytics Use Case	Typical Impact	Translated CO ₂ Reduction	Sources
Improved short-term load forecasting	10–20% forecasting error reduction	up to 10% in fossil-heavy grids	[175, 176]
Demand-response (residential and C&I)	5–15% household energy reduction	3–9% CO ₂ reduction; 150–300 kg CO ₂ per household annually	[178, 179]
Behavioural efficiency enabled by consumption feedback	4–7% household energy reduction	100–250 kg CO ₂ savings per household annually	[180]
Energy theft and anomaly detection	30–60% detection improvement	12–28 kg CO ₂ saved per connection per month	[181, 182]
EV smart charging	20–40% reduction in peak charging emissions	0.1–2.5 t CO ₂ saved per vehicle annually	[183, 184]
ESS optimization & PV self-consumption	18.8–20.8% reduction in grid imports during evening peaks	200–450 kg CO ₂ per household annually	[185]
Outage analytics & accelerated restoration	20–60% faster localization	5–12 kg CO ₂ savings per event per customer	[186]
System-level integration (combined effects)	Cross-domain effects from forecasting, DR, theft detection, EV/ESS	30% annual distribution-level emissions reduction	[187]

Table 5

Role of data-driven analytics in supporting net-zero objectives.

Analytics Domain	Primary Function	Net-Zero Contribution
Demand Forecasting	Predict short- and long-term energy consumption	Enables optimal scheduling of low-carbon generation and reduces reliance on reserves
Electricity Theft Detection	Identify and mitigate non-technical losses	Improves grid efficiency, reducing generation needs and associated emissions
Outage Detection	Locate and respond to faults quickly	Enhances resilience, allowing more renewable integration with fewer service disruptions
Anomaly Detection	Detect equipment issues and cyber-physical threats	Prevents energy loss and system inefficiencies that indirectly increase emissions
Topology Identification	Map actual network configuration and connectivity	Enables dynamic control, load balancing, and DER coordination for decarbonized grids
EV & ESS Integration	Optimize charging/discharging and grid interaction	Aligns flexible loads with renewable availability, reducing peak fossil generation

However, their real-world implementation is often hindered by outdated geographic information, incomplete network topology records, and limited forecasting of future mobility trends. Without comprehensive integration between transportation planning and electricity distribution systems, charging infrastructure risks being either underutilized or overly burdensome to the grid.

The integration of EVs also requires greater interoperability and standardization of communication protocols between vehicles, chargers, and distribution management systems. The current lack of unified standards restricts the deployment of advanced coordination mechanisms, such as V2G services, which could otherwise transform EVs from passive loads into active grid assets. This gap highlights the need for a systems-level approach to network intelligence, one that can harmonize control logic across different domains while maintaining cybersecurity and data privacy.

The decarbonisation of transport through EVs presents an unprecedented opportunity to reshape energy and mobility systems simultaneously. However, it also demands a new level of intelligence in distribution network operation, one capable of dealing with scale, heterogeneity, and uncertainty. Network intelligence must evolve from centralized monitoring to dynamic, context-aware decision-making, supporting a resilient and carbon-free transport-energy ecosystem.

4.4. Network intelligence and digital twin

Digital twin technology is emerging as a transformative asset in the development of intelligent distribution networks. By creating dynamic, virtual replicas of physical infrastructure, digital twins allow DNOs to simulate, monitor, and optimize grid operations in real time. As discussed in Section 3, managing modern distribution systems, characterized by high penetrations of DERs, EVs, and bidirectional flows, requires more than traditional supervisory tools. Digital twins meet this need by integrating data from smart meters, weather systems, and operational devices to model the behavior of grid components under various scenarios. These simulations enable predictive maintenance, adaptive control strategies, and scenario planning, which are vital for improving reliability, reducing operational costs, and enhancing grid flexibility [188].

In practice, digital twin platforms offer substantial potential for operational foresight and coordination. For example, they can forecast equipment degradation, test the impact of distributed energy integration, or support the optimal dispatch of energy storage during peak periods. In Australia, the Victorian government has already established a dedicated digital twin initiative to consolidate cross-sector datasets, offering a blueprint for broader adoption within the energy sector. When

combined with smart meter data and real-time operational metrics, such platforms can significantly enhance decision-making at both the asset and system level [189]. However, this expanded intelligence introduces computational and architectural challenges. Managing large, multi-source datasets requires advanced analytics, scalable infrastructure, and real-time processing capabilities. Moreover, ensuring the fidelity and accuracy of the digital twin model demands continuous validation and synchronization with live system conditions [190].

Looking forward, the widespread implementation of digital twins in distribution networks will depend on the development of interoperable frameworks and open data standards. Their full value lies in bridging the gap between real-world operations and advanced analytics, providing utilities with the situational awareness and agility required in an increasingly complex and low-carbon energy environment. As the grid continues to evolve, digital twins will become a cornerstone of network intelligence, supporting adaptive, data-driven management at the core of future-ready power systems.

4.5. Network intelligence and fog computing

The modern distribution grid generates massive volumes of data from smart meters, sensors, EV chargers, and DERs. Processing this data in real time is critical for ensuring operational efficiency, especially in scenarios that demand rapid detection and response, such as load imbalances, equipment failures, and voltage anomalies. While cloud computing has traditionally supported data storage, historical analysis, and visualization, it is often constrained by latency, bandwidth usage, and centralization-related security risks. To address these limitations, fog computing has emerged as a complementary solution that decentralizes data processing by placing computation resources closer to the source of data generation [191].

By shifting analytics from centralized servers to edge devices, such as smart meters, gateways, or local substations, fog computing enables real-time decision-making without depending on continuous cloud connectivity. For instance, localized outage detection or EV charging coordination can be executed directly at the edge, improving system responsiveness and reducing data transmission overhead. As described in Section III, applications such as anomaly detection and topology identification benefit significantly from lower-latency analytics. The ability to act on data locally also enhances grid resilience, especially in emergency or disconnected conditions. Additionally, fog computing supports hierarchical control architectures, where preliminary processing occurs at the edge and aggregated results are transmitted to the cloud for long-term optimization and planning [192].

As DNOs expand their digital infrastructure, fog computing is expected to play a critical role in balancing centralized and decentralized intelligence. It offers a scalable framework for integrating diverse analytics workloads, accommodating real-time demands at the operational edge while maintaining the strategic oversight of cloud-based platforms. To realize its full potential, however, utilities must address challenges such as interoperability between fog and cloud systems, standardization of edge computing protocols, and cybersecurity at distributed nodes. Nonetheless, fog computing represents a foundational enabler of real-time network intelligence and is critical for the future of responsive, data-driven grid operation. Fig. 7 presents a conceptual architecture of intelligent distribution network operation, highlighting the data and control flows across device-level sensing, edge (fog) computing, centralized cloud analytics, and decision-making layers.

4.6. Data governance, privacy, and regulatory frameworks

The growing reliance on smart meter data for forecasting, anomaly detection, behavioural modelling, and DER optimization has intensified the need for robust data governance and privacy-preserving mechanisms across modern distribution networks. As analytics pipelines become more complex and data volumes expand, utilities must ensure that smart

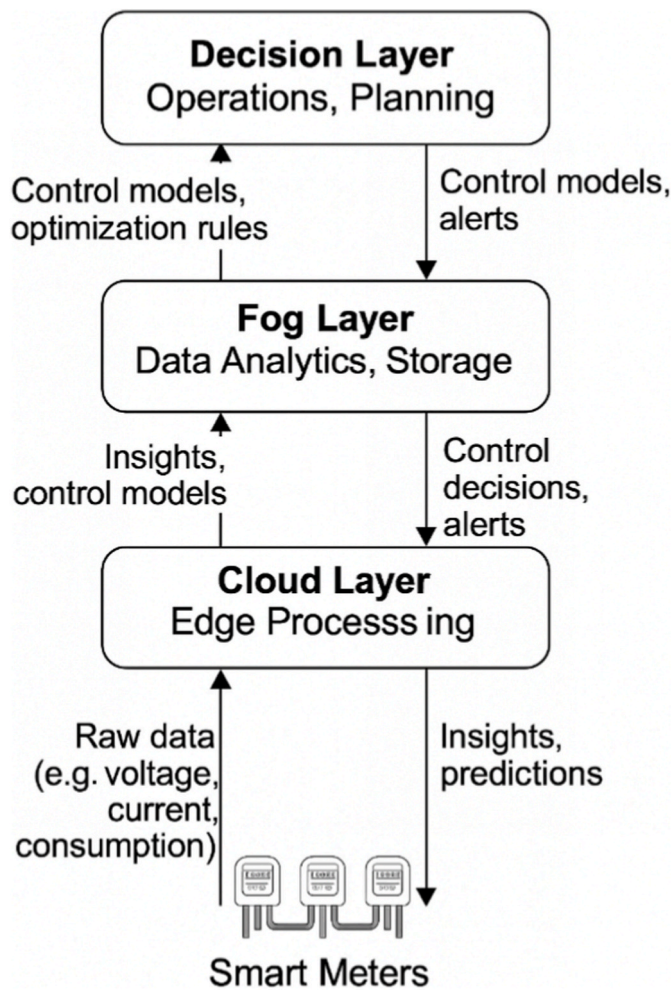


Fig. 7. Overview of architecture of Intelligent Data-Driven Distribution Network Operations.

meter data are collected, processed, and stored in ways that maintain confidentiality, integrity, and regulatory compliance. Recent work highlights that strong governance frameworks are critical for sustaining public trust and enabling secure, scalable data-driven energy operations [193,194].

Smart meter consumption data are increasingly recognized as sensitive because they can reveal occupancy patterns, appliance usage, behavioural routines, and even socio-economic characteristics. The European Union's General Data Protection Regulation (GDPR) explicitly classifies high-resolution consumption data as personal data when associated with identifiable households, imposing strict obligations related to consent, purpose limitation, and minimisation [195]. Comparable requirements exist in North America through guidelines such as NISTIR 7628-Rev.1 and in international standards like ISO/IEC 27557, which define privacy-enhanced processes for energy data analytics [196, 197]. These frameworks shape how DNOs justify data processing activities and restrict the repurposing of consumption data for secondary applications without explicit consumer agreement.

Data governance extends beyond privacy to encompass data quality, lineage, interoperability, access control, and auditability. Utilities are increasingly adopting structured governance models that incorporate data stewardship roles, metadata registries, and lifecycle documentation to manage the flow of smart meter data from acquisition to analytical output. Governance systems ensure that analytical models are trained on high-quality, well-documented data, thereby improving reproducibility and supporting regulatory audits [198]. Data-sharing agreements have

also become necessary where multiple actors contribute to or consume smart meter analytics, such as aggregators, DR service providers, and third-party analytics vendors. These agreements formalize responsibilities for data storage, access privileges, and security obligations across organisational boundaries.

Privacy-preserving analytics has emerged as a promising solution for deriving value from smart meter data without exposing sensitive information. Federated learning enables models to be trained collaboratively across distributed devices or local data silos while keeping raw data on premises. This approach has been successfully applied to load forecasting, energy theft detection, and behavioural modelling, demonstrating improved privacy protection without significantly reducing model accuracy [199,200]. Despite these advantages, federated learning introduces challenges such as communication overhead, vulnerability to poisoned model updates, and the need for adaptive aggregation techniques to handle heterogeneous client data. Addressing these challenges remains an active research area in energy data analytics.

Several complementary privacy-enhancing technologies (PETs) further strengthen smart meter data protection. Differential privacy adds calibrated noise to aggregated statistics or model outputs, enabling system-level insights while reducing re-identification risk [201]. Secure multi-party computation and homomorphic encryption provide cryptographic guarantees that allow collaborative computations over encrypted data, though their computational cost can limit adoption in latency-sensitive grid applications [202]. Recent studies demonstrate progress in reducing these computational burdens, making encrypted analytics increasingly feasible for selected smart grid tasks, such as billing verification and aggregated forecasting [203].

Standardisation efforts play a central role in ensuring secure, interoperable data exchange. Communication and data modelling standards such as IEC 61968/61970 (CIM), IEC 62056 (DLMS/COSEM), and IEEE 2030.5 support consistent and secure handling of smart meter data across heterogeneous platforms [204–206]. Cybersecurity standards including IEC 62443 and ISO/IEC 27019 define requirements for authentication, encryption, intrusion detection, and firmware integrity for energy systems, providing a structured defence-in-depth strategy for both centralised and edge-based analytics [207]. As the industry increasingly shifts toward edge computing, these standards help mitigate vulnerabilities associated with distributed intelligence.

Data sovereignty laws introduce further considerations, with several jurisdictions mandating that identifiable consumer data remain within national borders or be stored in accredited data centers. These requirements often lead to hybrid analytics architectures in which identifiable data are processed locally, while aggregated or anonymised insights are transmitted to cloud platforms for large-scale optimization [208,209]. Achieving compliance requires solutions such as anonymisation pipelines, tokenisation methods, or secure execution environments that protect user identity while preserving analytical utility.

Organisational readiness remains a major barrier to effective data governance implementation. Many utilities lack personnel trained in privacy engineering, data ethics, and cybersecurity. Developing interdisciplinary teams that combine legal, technical, and operational expertise is essential for responsible deployment of advanced analytics [210,211]. Furthermore, consumer acceptance depends on transparent communication about how smart meter data are used, stored, and protected, underscoring the role of clear privacy notices and participatory governance mechanisms.

In summary, the increasing sophistication of smart meter analytics requires equally mature frameworks for data governance, privacy protection, and regulatory compliance. By adopting federated and privacy-preserving analytics, aligning with international standards, strengthening governance infrastructures, and building organisational capacity, DNOs can leverage rich consumption data responsibly. These measures ensure that smart meter intelligence advances securely, ethically, and in alignment with societal expectations for digital trust.

5. Toward intelligent, data-centric distribution networks

5.1. Digital transformation of distribution network operations

The digital evolution of distribution networks is accelerating as high-resolution smart metering, advanced analytics, and increasingly diverse data streams converge into a unified operational ecosystem. Modern distribution system operators are transitioning from static control philosophies to adaptive, predictive, and learning-based operational regimes. This shift is driven by the growing ability to observe consumption patterns, equipment health, and network conditions at fine temporal and spatial scales. Data-driven intelligence is becoming a core architectural foundation for meeting rapidly rising expectations surrounding reliability, flexibility, and sustainability across complex energy systems.

5.2. System-level pressures from electrification and decarbonisation

The widespread electrification of transport, heating, and industrial activity, combined with the deepening penetration of rooftop solar and behind-the-meter storage, introduces new forms of variability and decentralization. Power flows now frequently reverse direction, voltage behavior is increasingly shaped by local consumer actions, and clusters of prosumers have measurable influence on feeder performance. These dynamics demand analytics that function across multiple timescales. Real-time inference is required for protection and stability, while medium-term forecasting guides planning and investment decisions. High-resolution smart meter data serve as the connective layer between these tasks, capturing interactions between distributed resources, occupant activities, and grid operational states.

5.3. Emerging computational architectures for grid intelligence

A new generation of analytical platforms is reshaping how smart meter data are interpreted and acted upon. Digital twins, built from continuously refreshed measurements and hybrid physical-data models, support scenario evaluation, predictive maintenance, and automated optimization. Their effectiveness depends directly on the fidelity and consistency of the underlying data streams. Fog and edge computing extend these capabilities by enabling localized processing and low-latency decision-making. This reduces communication burdens while empowering devices such as EV chargers, solar inverters, and home energy management systems to coordinate their behavior with feeder constraints. Together, these tools support a vision of distribution networks operating as adaptive ecosystems that learn from their own data.

5.4. Data quality, uncertainty, and model robustness challenges

Despite technical progress, fundamental data challenges remain unresolved. Variability in metering hardware, inconsistent sampling rates, missing values, and timestamp irregularities can impair analytical models and produce misleading operational insights. The introduction of high-resolution meters intensifies these issues by increasing data volume and exposing more noise and irregularities. Machine learning models also face difficulties when applied to data that are non-stationary, context-dependent, and influenced by human behavior. Ensuring that these models perform reliably outside controlled pilot environments is a major research challenge and a prerequisite for large-scale deployment.

5.5. Interpretability, transparency, and institutional trust

As data-driven tools assume greater influence over automated control, forecasting pipelines, and reliability-related analytics, the demand for interpretability grows. Operators and regulators must be able to understand and verify the reasoning behind model outputs, particularly in safety-critical or customer-facing applications. Techniques that

provide uncertainty quantification, scenario-based explanations, and feature attributions will be essential for establishing trust. This is especially relevant for deep learning models, which often achieve high predictive accuracy but offer limited transparency regarding their internal reasoning.

5.6. Governance, standardization, and data access frameworks

Institutional and governance factors will significantly shape the future of intelligent distribution systems. Standardized data formats, interoperable device ecosystems, and clear privacy protections are necessary for enabling data sharing and reuse. Utilities must develop secure data pipelines, formalize access control procedures, and align with regulatory requirements that define acceptable uses of customer data. Emerging concepts such as national or regional data hubs offer promising directions for coordinated data access, although they require robust cybersecurity protections, policy alignment, and broad stakeholder engagement to realize their potential.

5.7. Organisational readiness and workforce capability

Developing and maintaining intelligent distribution networks will require skillsets that combine power engineering, data science, cybersecurity, and regulatory knowledge. Utilities need analysts who can interpret complex data streams, operators who can integrate advanced decision-support tools into workflow, and planners who understand both technical constraints and digital opportunities. Workforce development therefore becomes an essential enabler of the broader digital transition.

5.8. Strategic role of smart meter intelligence in a net-zero future

Smart meter intelligence provides a powerful pathway toward sustainable and electrification-ready distribution systems. Data-driven tools can reduce operational carbon intensity, support rapid integration of DERs, and empower consumers to participate in local flexibility programs. Investments in data infrastructure, analytical capability, and governance frameworks will allow utilities to shift from reactive management to proactive, adaptive, and climate-aligned operations. Within this broader transformation, smart meter intelligence functions not only as an efficiency tool but also as a strategic instrument for system-wide decarbonisation and long-term energy resilience.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] Al Khafaf N, Wang J, Jalili M, Sokolowski P. A deep learning approach for detecting distributed generation in residential customers. *Sustain Energy Grids Netw* 2023;33:100966. <https://doi.org/10.1016/j.segan.2022.100966>.
- [2] Alahakoon D, Yu X. Smart electricity meter data intelligence for future energy systems: a survey. *IEEE Trans Ind Inf* 2016;12(1):425–36. <https://doi.org/10.1109/TII.2015.2414355>.
- [3] Alahakoon D, Yu X. Smart electricity meter data intelligence for future energy systems: a survey. *IEEE Trans Ind Inf* 2015;12(1):425–36.
- [4] Bhattarai BP, et al. Big data analytics in smart grids: state-of-the-art, challenges, opportunities, and future directions. *IET Smart Grid* 2019;2(2):141–54.
- [5] Connor P, Axon C, Xenias D, Balta-Ozkan N. Sources of risk and uncertainty in UK smart grid deployment: an expert stakeholder analysis. *Energy* 2018;161:1–9.
- [6] Rossi B, Chren S. Smart grids data analysis: a systematic mapping study. *IEEE Trans Ind Inf* 2020;16(6):3619–39. <https://doi.org/10.1109/TII.2019.2954098>.
- [7] Bhela S, Kekatos V, Veeramachaneni S. Enhancing observability in distribution grids using smart meter data. *IEEE Trans Smart Grid* 2018;9(6):5953–61. <https://doi.org/10.1109/TSG.2017.2699939>.
- [8] Jin N, Flach P, Wilcox T, Sellman R, Thumim J, Knobbe A. Subgroup Discovery in smart electricity meter data. *IEEE Trans Ind Inf* 2014;10(2):1327–36. <https://doi.org/10.1109/TII.2014.2311968>.
- [9] C. M. R. Do Carmo and T. H. Christensen, "Cluster analysis of residential heat load profiles and the role of technical and household characteristics," *Energy Build*, vol. 125, pp. 171–180, doi: 10.1016/j.enbuild.2016.04.079.
- [10] Haben S, Singleton C, Grindrod P. Analysis and clustering of residential customers energy behavioral demand using smart meter data. *IEEE Trans Smart Grid* 2016;7(1):136–44. <https://doi.org/10.1109/TSG.2015.2409786>.
- [11] Al Khafaf N, et al. Impact of battery storage on residential energy consumption: an Australian case study based on smart meter data. *Renew Energy* 2022;182:390–400. <https://doi.org/10.1016/j.renene.2021.10.005>.
- [12] Al Khafaf N, Jalili M, Sokolowski P. A novel clustering index to Find optimal clusters size with application to segmentation of energy consumers. *IEEE Trans Ind Inf* 2021;17(1):346–55. <https://doi.org/10.1109/TII.2020.2987320>.
- [13] Jiang G, Chen Y, Wang Z, Powell K, Billings B, Chen J. A deep learning-based Bayesian framework for high-resolution calibration of building energy models. *Energy Build* 2024;323:114755. <https://doi.org/10.1016/j.enbuild.2024.114755>.
- [14] Avancini DB, Rodrigues JJPC, Martins SGB, Rabêlo RAL, Al-Muhtadi J, Solic P. Energy meters evolution in smart grids: a review. *J Clean Prod* 2019;217:702–15. <https://doi.org/10.1016/j.jclepro.2019.01.229>.
- [15] Rodrigues Junior WL, Borges FAS, Veloso AfdS, Rabêlo RdAL, Rodrigues JJPC. Low voltage smart meter for monitoring of power quality disturbances applied in smart grid. *Measurement : journal of the International Measurement Confederation* 2019;147(C):106890. <https://doi.org/10.1016/j.measurement.2019.106890>.
- [16] Stegner C, Glaß O, Beikircher T. Comparing smart metered, residential power demand with standard load profiles. *Sustain Energy Grids Netw* 2019;20:100248. <https://doi.org/10.1016/j.segan.2019.100248>.
- [17] Chen X, Cai S, Smeaton AF. Back-filling missing data when predicting Domestic electricity consumption from smart meter data. 2024. <https://doi.org/10.48550/arxiv.2412.03574>.
- [18] Claeys R, Cleenwerck R, Knockaert J, Desmet J. Stochastic generation of residential load profiles with realistic variability based on wavelet-decomposed smart meter data. *Appl Energy* 2023;350:121750. <https://doi.org/10.1016/j.apenergy.2023.121750>.
- [19] Boopathy P, et al. Deep learning for intelligent demand response and smart grids: a comprehensive survey. *Computer science review* 2024;51:100617. <https://doi.org/10.1016/j.cosrev.2024.100617>.
- [20] Sousa ELD, et al. Development a low-cost Wireless smart meter with power quality measurement for smart grid applications. *Sensors (Basel, Switzerland)* 2023;23(16):7210. <https://doi.org/10.3390/s23167210>.
- [21] Sun M, Zhang T, Wang Y, Strbac G, Kang C. Using bayesian deep learning to capture uncertainty for residential net load forecasting. *IEEE Trans Power Syst* 2020;35(1):188–201. <https://doi.org/10.1109/TPWRS.2019.2924294>.
- [22] Yang Y, Hong W, Li S. Deep ensemble learning based probabilistic load forecasting in smart grids. *Energy (Oxford)* 2019;189:116324. <https://doi.org/10.1016/j.energy.2019.116324>.
- [23] Ye X, Zhang Z, Qiu Y. Review of application of high frequency smart meter data in energy economics and policy research. *Frontiers in sustainable energy policy* 2023;2. <https://doi.org/10.3389/fsuep.2023.1171093>.
- [24] Ahsan F, et al. Data-driven next-generation smart grid towards sustainable energy evolution: techniques and technology review. *Protection and control of modern power systems* 2023;8(1):43. <https://doi.org/10.1186/s41601-023-00319-5>.
- [25] Yildiz B, Bilbao Ji, Dore J, Sproul AB. Recent advances in the analysis of residential electricity consumption and applications of smart meter data. *Appl Energy* 2017;208:402–27.
- [26] Wang Y, Chen Q, Hong T, Kang C. Review of smart meter data analytics: applications, methodologies, and challenges. *IEEE Trans Smart Grid* 2018;10(3):3125–48.
- [27] Lopez J, Rubio JE, Alcaraz C. A resilient architecture for the smart grid. *IEEE Trans Ind Inf* 2018;14(8):3745–53.
- [28] Yang Y, Li W, Gulliver TA, Li S. Bayesian deep learning-based probabilistic load forecasting in smart grids. *IEEE Trans Ind Inf* 2019;16(7):4703–13.
- [29] Wen L, Zhou K, Yang S, Li L. Compression of smart meter big data: a survey. *Renew Sustain Energy Rev* 2018;91:59–69.
- [30] Chowdhury D, Sarkar M, Haider MZ, Alam T. Zone wise hourly load prediction using regression decision tree model. In: 2018 international Conference on innovation in engineering and technology (ICIET). IEEE; 2018. p. 1–6.
- [31] Zhang G, Guo J. A novel method for hourly electricity demand forecasting. *IEEE Trans Power Syst* 2019;35(2):1351–63.
- [32] Jiang P, Li R, Liu N, Gao Y. A novel composite electricity demand forecasting framework by data processing and optimized support vector machine. *Appl Energy* 2020;260:114243.
- [33] Ouyang T, He Y, Li H, Sun Z, Baek S. Modeling and forecasting short-term power load with copula model and deep belief network. *IEEE Trans Emerg Top Comput Intell* 2019;3(2):127–36.
- [34] Pelka P, Dudek G. Pattern-based long short-term memory for Mid-term electrical load forecasting. 2020. *arXiv preprint arXiv:2004.11834*.
- [35] Kazemzadeh M-R, Amjadian A, Amraee T. A hybrid data mining driven algorithm for long term electric peak load and energy demand forecasting. *Energy* 2020: 117948.
- [36] Jiang H. A short-term and high-resolution system load forecasting approach using support vector regression with hybrid parameters optimization, vols. 1949–3053. Golden, CO (United States): National Renewable Energy Lab.(NREL); 2017.
- [37] Sajjad M, et al. A novel CNN-GRU-based hybrid approach for short-term residential load forecasting. *IEEE Access* 2020;8:143759–68.
- [38] Rahman A, Srikumar V, Smith AD. Predicting electricity consumption for commercial and residential buildings using deep recurrent neural networks. *Appl Energy* 2018;212:372–85.
- [39] Tan M, Yuan S, Li S, Su Y, Li H, He F. Ultra-Short-Term Industrial Power Demand Forecasting Using LSTM Based Hybrid Ensemble Learning. *IEEE Trans Power Syst* 2020;35(4):2937–48. <https://doi.org/10.1109/TPWRS.2019.2963109>.
- [40] Song H, Qin AK, Salim FD. Multi-resolution selective ensemble extreme learning machine for electricity consumption prediction. In: International Conference on neural information processing. Springer; 2017. p. 600–9.
- [41] Cao Z, Wan C, Zhang Z, Li F, Song Y. Hybrid ensemble deep learning for deterministic and probabilistic low-voltage load forecasting. *IEEE Trans Power Syst* 2019;35(3):1881–97.
- [42] Song H, Qin A, Salim FD. Evolutionary model construction for electricity consumption prediction. *Neural Comput Appl* 2020;32(16):12155–72.
- [43] Eseye AT, Lehtonen M, Tukka T, Uimonen S, Millar RJ. Machine learning based integrated feature selection approach for improved electricity demand forecasting in decentralized energy systems. *IEEE Access* 2019;7:91463–75.
- [44] Hsiao Y-H. Household electricity demand forecast based on context information and user daily schedule analysis from meter data. *IEEE Trans Ind Inf* 2014;11(1):33–43.
- [45] Song H, Qin AK, Salim FD. Multivariate electricity consumption prediction with extreme learning machine. In: 2016 international joint Conference on neural networks (IJCNN). IEEE; 2016. p. 2313–20.
- [46] Song H, Qin AK, Salim FD. Evolutionary multi-objective ensemble learning for multivariate electricity consumption prediction. In: 2018 international joint Conference on neural networks (IJCNN). IEEE; 2018. p. 1–8.
- [47] Abedinia O, Amjadi N, Zareipour H. A new feature selection technique for load and price forecast of electrical power systems. *IEEE Trans Power Syst* 2016;32(1):62–74.
- [48] Bento P, Pombo J, Calado M, Mariano S. Optimization of neural network with wavelet transform and improved data selection using bat algorithm for short-term load forecasting. *Neurocomputing* 2019;358:53–71.
- [49] Dong Z, Tian Z, Lv S. A short-term power load forecasting system based on data decomposition, deep learning and weighted linear error correction with feedback mechanism. *Appl Soft Comput* 2024;162:111863. <https://doi.org/10.1016/j.asoc.2024.111863>.
- [50] Al-Ja'afreh MAA, Mokryani G, Amjad B. An enhanced CNN-LSTM based multi-stage framework for PV and load short-term forecasting: DSO scenarios. *Energy Rep* 2023;10:1387–408. <https://doi.org/10.1016/j.egy.2023.08.003>.
- [51] Zulfiqar M, Kamran M, Rasheed MB, Alquthami T, Milyani AH. A hybrid framework for short term load forecasting with a novel feature engineering and adaptive grasshopper optimization in smart grid. *Appl Energy* 2023;338:120829. <https://doi.org/10.1016/j.apenergy.2023.120829>.
- [52] Peplinski M, et al. A machine learning framework to estimate residential electricity demand based on smart meter electricity, climate, building characteristics, and socioeconomic datasets. *Appl Energy* 2024;357:122413.
- [53] Kim H, Park S, Kim S. Time-series clustering and forecasting household electricity demand using smart meter data. *Energy Rep* 2023;9:4111–21.
- [54] Mbey CF, et al. Solar photovoltaic generation and electrical demand forecasting using multi-objective deep learning model for smart grid systems. *Cogent Eng* 2024;11(1):2340302.
- [55] Yao D, Wen M, Liang X, Fu Z, Zhang K, Yang B. Energy theft detection with energy privacy Preservation in the smart grid. *IEEE Internet Things J* 2019;6(5):7659–69. <https://doi.org/10.1109/JIOT.2019.2903312>.
- [56] Punmiya R, Choe S. Energy theft detection using gradient boosting theft detector with feature engineering-based preprocessing. *IEEE Trans Smart Grid* 2019;10(2):2326–9. <https://doi.org/10.1109/TSG.2019.2892595>.
- [57] Liu Y, Liu T, Sun H, Zhang K, Liu P. Hidden electricity theft by Exploiting multiple-pricing scheme in smart grids. *IEEE Trans Inf Forensics Secur* 2020;15:2453–68. <https://doi.org/10.1109/TIFS.2020.2965276>.
- [58] Tariq M, Poor HV. Electricity theft detection and localization in grid-Tied Microgrids. *IEEE Trans Smart Grid* 2018;9(3):1920–9. <https://doi.org/10.1109/TSG.2016.2602660>.
- [59] Ibrahim MI, Nabil M, Fouda MM, Mahmoud M, Alasmay W, Alsolami F. Efficient privacy-preserving electricity theft detection with dynamic billing and load

- monitoring for AMI networks. *IEEE Internet Things J* 2020. <https://doi.org/10.1109/JIOT.2020.3026692>. 0–1.
- [60] Salinas SA, Li P. Privacy-preserving energy theft detection in Microgrids: a state estimation approach. *IEEE Trans Power Syst* 2016;31(2):883–94. <https://doi.org/10.1109/TPWRS.2015.2406311>.
- [61] Li W, Logenthiran T, Phan V, Woo WL. A novel smart energy theft system (SETS) for IoT-based smart home. *IEEE Internet Things J* 2019;6(3):5531–9. <https://doi.org/10.1109/JIOT.2019.2903281>.
- [62] Xia X, Xiao Y, Liang W. SAI: a suspicion assessment-based inspection algorithm to detect Malicious users in smart grid. *IEEE Trans Inf Forensics Secur* 2020;15: 361–74. <https://doi.org/10.1109/TIFS.2019.2921232>.
- [63] Massafiero P, Martino JMD, Fernández A. Fraud detection in electric power distribution: an approach that maximizes the economic return. *IEEE Trans Power Syst* 2020;35(1):703–10. <https://doi.org/10.1109/TPWRS.2019.2928276>.
- [64] Zheng K, Chen Q, Wang Y, Kang C, Xia Q. A novel combined data-driven approach for electricity theft detection. *IEEE Trans Ind Inf* 2019;15(3):1809–19. <https://doi.org/10.1109/TII.2018.2873814>.
- [65] Gao Y, Foggo B, Yu N. A physically Inspired data-driven model for electricity theft detection with smart meter data. *IEEE Trans Ind Inf* 2019;15(9):5076–88. <https://doi.org/10.1109/TII.2019.2898171>.
- [66] He Y, Mendis GJ, Wei J. Real-time detection of false data injection attacks in smart grid: a deep learning-based intelligent mechanism. *IEEE Trans Smart Grid* 2017;8(5):2505–16. <https://doi.org/10.1109/TSG.2017.2703842>.
- [67] Ismail M, Shaaban MF, Naidu M, Serpedin E. Deep learning detection of electricity theft cyber-attacks in renewable distributed generation. *IEEE Trans Smart Grid* 2020;11(4):3428–37. <https://doi.org/10.1109/TSG.2020.2973681>.
- [68] Zheng Z, Yang Y, Niu X, Dai H, Zhou Y. Wide and deep convolutional neural networks for electricity-theft detection to secure smart grids. *IEEE Trans Ind Inf* 2018;14(4):1606–15. <https://doi.org/10.1109/TII.2017.2785963>.
- [69] Jindal A, Dua A, Kaur K, Singh M, Kumar N, Mishra S. Decision tree and SVM-based data analytics for theft detection in smart grid. *IEEE Trans Ind Inf* 2016;12(3):1005–16. <https://doi.org/10.1109/TII.2016.2543145>.
- [70] Zhou Y, Chen X, Zomaya AY, Wang L, Hu S. A dynamic programming algorithm for leveraging probabilistic detection of energy theft in smart home. *IEEE Trans Emerg Top Comput* 2015;3(4):502–13. <https://doi.org/10.1109/TETC.2015.2484841>.
- [71] Jokar P, Arianpoo N, Leung VCM. Electricity theft detection in AMI using customers' consumption patterns. *IEEE Trans Smart Grid* 2016;7(1):216–26. <https://doi.org/10.1109/TSG.2015.2425222>.
- [72] Zanetti M, Jamhour E, Pellenz M, Penna M, Zambenedetti V, Chueiri I. A Tunable Fraud detection system for advanced metering infrastructure using short-Lived patterns. *IEEE Trans Smart Grid* 2019;10(1):830–40. <https://doi.org/10.1109/TSG.2017.2753738>.
- [73] Hasnat MA, Anand H, Tootkaboni M, Alemazkoo N. Spatio-temporal graph attention network-based detection of FDIA from smart meter data at geographically hierarchical levels. *Elec Power Syst Res* 2025;238:111149. <https://doi.org/10.1016/j.epsr.2024.111149>.
- [74] Wang X, Xie H, Tang L, Chen C, Bie Z. Decentralized privacy-preserving electricity theft detection for distribution system operators. *IEEE Trans Smart Grid* 2024;15(2):2179–90. <https://doi.org/10.1109/TSG.2023.3313771>.
- [75] Mbey CF, Bikai J, Yem Souhe FG, Foba Kakeu, Boum AT. Electricity theft detection in a smart grid using hybrid deep learning-based data analysis technique. *J Electr Comput Eng* 2024;2024(1):6225510. <https://doi.org/10.1155/2024/6225510>.
- [76] Xia X, et al. ETD-ConvLSTM: a deep learning approach for electricity theft detection in smart grids. *IEEE Trans Inf Forensics Secur* 2023;18:2553–66.
- [77] Rovatos G, Jiang X, Domínguez-García AD, Veeravalli VV. Statistical power system line outage detection under transient dynamics. *IEEE Trans Signal Process* 2017;65(11):2787–97.
- [78] Samudrala AN, Amini MH, Kar S, Blum RS. Sensor placement for outage Identifiability in power distribution networks. *IEEE Trans Smart Grid* 2019;11(3): 1996–2013.
- [79] Sevilian RA, Zhao Y, Rajagopal R, Goldsmith A, Poor HV. Outage detection using load and line flow measurements in power distribution systems. *IEEE Trans Power Syst* 2017;33(2):2053–69.
- [80] Nie S, Ding L, Li W. Multiple line-outage detection in power system with load stochastic perturbations. *IEEE Trans Circuits Syst II, Exp Briefs* 2019;67(10): 1994–8. <https://doi.org/10.1109/TCSII.2019.2940095>.
- [81] Zhao Y, Chen J, Poor HV. A learning-to-infer method for real-time power grid multi-line outage identification. *IEEE Trans Smart Grid* 2019;11(1):555–64.
- [82] Yuan Y, Dehghanpour K, Bu F, Wang Z. Outage detection in partially observable distribution systems using smart meters and generative adversarial networks. *IEEE Trans Smart Grid* 2020;11(6):5418–30.
- [83] Black J, Hoffman A, Hong T, Roberts J, Wang P. Weather data for energy analytics: from modeling outages and reliability indices to simulating distributed photovoltaic fleets. *IEEE Power Energy Mag* 2018;16(3):43–53.
- [84] Bashkari MS, Sami A, Rastegar M. Outage cause detection in power distribution systems based on data mining. *IEEE Trans Ind Inform* 2021;17(1):640–9. <https://doi.org/10.1109/TII.2020.2966505>.
- [85] Chung H-M, Li W-T, Yuen C, Chung W-H, Zhang Y, Wen C-K. Local cyber-physical attack for masking line outage and topology attack in smart grid. *IEEE Trans Smart Grid* 2018;10(4):4577–88.
- [86] Bauman K, Tuzhilin A, Zaczynski R. Using sensor sensors for detecting emergency events: a case of power outages in the electrical utility industry. *ACM Transactions on Management Information Systems (TMIS)* 2017;8(2–3):1–20.
- [87] Samudrala AN, Amini MH, Kar S, Blum RS. Distributed outage detection in power distribution networks. *IEEE Trans Smart Grid* 2020;11(6):5124–37.
- [88] Ahmed A, et al. Multiple power line outage detection in smart grids: probabilistic bayesian approach. *IEEE Access* 2017;6:10650–61.
- [89] Yang F, Tan J, Song J, Han Z. Block-wise compressive sensing based multiple line outage detection for smart grid. *IEEE Access* 2018;6:50984–93.
- [90] Babakmehr M, Simoes MG, Al-Durra A, Harirchi F, Han Q. Application of compressive sensing for distributed and structured power line outage detection in smart grids. In: 2015 American control Conference (ACC). IEEE; 2015. p. 3682–9.
- [91] Joseph S, Jasmin E. Stream computing framework for outage detection in smart grid. In: 2015 international Conference on power, instrumentation, control and computing (PICC). IEEE; 2015. p. 1–5.
- [92] Yuan Y, et al. Outage detection in partially observable distribution systems using smart meters and generative adversarial networks. *IEEE Trans Smart Grid* 2020; 11(6):5418–26.
- [93] Liao Y, et al. Quick line outage identification in urban distribution grids via smart meters. *CSEE J Power Energy Syst* 2022;8(4):1074–82.
- [94] He X, Chu L, Qiu RC, Ai Q, Ling Z, Zhang J. Invisible units detection and estimation based on random matrix theory. *IEEE Trans Power Syst* 2019;35(3): 1846–55.
- [95] Huang S-C, Lo Y-L, Lu C-N. Non-technical loss detection using state estimation and analysis of variance. *IEEE Trans Power Syst* 2013;28(3):2959–66.
- [96] Long H, Chen C, Gu W, Xie J, Wang Z, Li G. A data-driven combined algorithm for abnormal power loss detection in the distribution network. *IEEE Access* 2020;8: 24675–86.
- [97] Passerini F, Tonello AM. Smart grid monitoring using power line modems: anomaly detection and localization. *IEEE Trans Smart Grid* 2019;10(6):6178–86.
- [98] Shi X, Qiu R, Ling Z, Yang F, Yang H, He X. Spatio-temporal correlation analysis of online monitoring data for anomaly detection and location in distribution networks. *IEEE Trans Smart Grid* 2019;11(2):995–1006.
- [99] Lo Y-L, Huang S-C, Lu C-N. Transformational benefits of AMI data in transformer load modeling and management. *IEEE Trans Power Deliv* 2013;29(2):742–50.
- [100] Srikantha P, Liu J, Samarabandu J. A novel distributed and Stealthy attack on active distribution networks and a mitigation strategy. *IEEE Trans Ind Inf* 2019;16(2):823–31.
- [101] Liu X, Che L, Gao K, Li Z. Power system intra-interval operational security under false data injection attacks. *IEEE Trans Ind Inf* 2019;16(8):4997–5008.
- [102] Singh SK, Khanna K, Bose R, Panigrahi BK, Joshi A. Joint-transformation-based detection of false data injection attacks in smart grid. *IEEE Trans Ind Inf* 2017;14(1):89–97.
- [103] Anwar A, Mahmood AN, Tari Z. Ensuring data integrity of OPF module and energy database by detecting changes in power flow patterns in smart grids. *IEEE Trans Ind Inf* 2017;13(6):3299–311.
- [104] Cui M, Wang J, Yue M. Machine learning-based anomaly detection for load forecasting under cyberattacks. *IEEE Trans Smart Grid* 2019;10(5):5724–34.
- [105] Shi X, Qiu R, Mi T, He X, Zhu Y. Adversarial feature learning of online monitoring data for operational risk assessment in distribution networks. *IEEE Trans Power Syst* 2019;35(2):975–85.
- [106] Jamei M, et al. Anomaly detection using optimally placed μPMU sensors in distribution grids. *IEEE Trans Power Syst* 2017;33(4):3611–23.
- [107] Cui M, Wang J, Florita AR, Zhang Y. Generalized graph Laplacian based anomaly detection for spatiotemporal microPMU data. *IEEE Trans Power Syst* 2019;34(5): 3960–3.
- [108] Hernández Á, et al. Detection of anomalies in daily activities using data from smart meters. *Sensors* 2024;24(2):515.
- [109] Zhou Z, et al. Two unsupervised learning algorithms for detecting abnormal inactivity within a household based on smart meter data. *Expert Syst Appl* 2023; 230:120565.
- [110] Alkuwari AN, et al. Anomaly detection on smart grids with optimized convolutional long short-term memory model. *IEEE Access* 2025;13:40399–415.
- [111] Farooq A, et al. Securing the green grid: a data anomaly detection method for mitigating cyberattacks on smart meter measurements. *International Journal of Critical Infrastructure Protection* 2024;46:100694.
- [112] Moghaddass R, Wang J. A hierarchical framework for smart grid anomaly detection using large-scale smart meter data. *IEEE Trans Smart Grid* 2017;9(6): 5820–30.
- [113] Cunha VC, Freitas W, Trindade FCL, Santos S. Automated Determination of topology and line parameters in low voltage systems using smart meters measurements. *IEEE Trans Smart Grid* 2020;11(6):5028–38. <https://doi.org/10.1109/TSG.2020.3004096>.
- [114] Mokhtar M, et al. Automating the verification of the low voltage network cables and topologies. *IEEE Trans Smart Grid* 2020;11(2):1657–66. <https://doi.org/10.1109/TSG.2019.2941722>.
- [115] Si F, Han Y, Wang J, Zhao Q. Connectivity verification in distribution systems using smart meter voltage analytics: a cloud-edge collaboration approach. *IEEE Trans Ind Inf* 2021;17(6):3929–39. <https://doi.org/10.1109/TII.2020.3012138>.
- [116] Guzman A, Arguello A, Quiros-Tortos J, Valverde G. Processing and correction of secondary system models in geographic information systems. *IEEE Trans Ind Inf* 2019;15(6):3482–91. <https://doi.org/10.1109/TII.2018.2876356>.
- [117] Al Khafaf N, Song H, McGrath B, Jalili M. Identification of low voltage distribution transformer–customer connectivity based on unsupervised learning. *Energy Rep* 2023;9:72–9. <https://doi.org/10.1016/j.egy.2022.12.104>.
- [118] Tennakoon RN, Kulatilaka AT, Giri R, Khafaf NA, Meegapola L, Jalili M. Prediction of customer connectivity to distribution transformer via application of machine learning techniques to smart-meter data. In: 2020 Australasian

- Universities power engineering Conference (AUPEC); 29 Nov.-2 Dec. 2020 2020. p. 1–6.
- [119] Jiao F, et al. Topology identification method for low-voltage distribution node networks based on density clustering using smart meter real-time measurement data. *IEEE Access* 2024;12:83600–15.
- [120] Pengwah AB, et al. Topology identification of distribution networks with partial smart meter coverage. *IEEE Trans Power Deliv* 2024;39(2):992–1004.
- [121] Flynn C, et al. An improved algorithm for topology identification of distribution networks using smart meter data and its application for fault detection. *IEEE Trans Smart Grid* 2023;14(5):3850–61.
- [122] Xu Z, et al. Distribution network topology identification using smart meter data and considering the same-bus-different-feeder condition. *IEEE Trans Ind Inf* 2023;19(8):8600–12.
- [123] García S, et al. Phase topology identification in low-voltage distribution networks: a bayesian approach. *Electrical Power and Energy Systems* 2023;144:108525.
- [124] Ahmed M, et al. Effects of household battery systems on LV residential feeder voltage management. *IEEE Trans Power Deliv* 2022;37(6). <https://doi.org/10.1109/TPWRD.2022.3176099>. 0–1.
- [125] Moradi Amani A, et al. Technology balancing for reliable EV uptake in distribution grids: an Australian case study. *Renew Energy* 2023;206:939–48. <https://doi.org/10.1016/j.renene.2023.02.056>.
- [126] Wang S, Bi S, Zhang YA. Reinforcement learning for real-time pricing and scheduling control in EV charging stations. *IEEE Trans Ind Inform* 2021;17(2): 849–59. <https://doi.org/10.1109/TII.2019.2950809>.
- [127] Yang J, Wang W, Ma K, Yang B, Dou C. Optimization and self-adaptive dispatching strategy for multiple shared battery stations of electric vehicles. *IEEE Trans Ind Inform* 2021;17(2):1363–74. <https://doi.org/10.1109/TII.2020.2983393>.
- [128] Shen J, Wang L, Zhang J. Integrated scheduling strategy for private electric vehicles and electric Taxis. *IEEE Trans Ind Inform* 2021;17(3):1637–47. <https://doi.org/10.1109/TII.2020.2993239>.
- [129] Chekired DA, Khoukhi L. Smart grid solution for charging and discharging services based on cloud computing scheduling. *IEEE Trans Ind Inf* 2017;13(6): 3312–21.
- [130] Procopiou AT, Quirós-Tortós J, Ochoa LF. HPC-based probabilistic analysis of LV networks with EVs: impacts and control. *IEEE Trans Smart Grid* 2016;8(3): 1479–87.
- [131] Parastvand H, Moghaddam V, Bass O, Masoum MA, Chapman A, Lachowicz S. A graph automorphic approach for placement and sizing of charging stations in EV network considering traffic. *IEEE Trans Smart Grid* 2020;11(5):4190–200.
- [132] Hashemian SN, Latify MA, Yousefi GR. PEV fast-charging station sizing and placement in coupled transportation-distribution networks considering power line conditioning capability. *IEEE Trans Smart Grid* 2020;11(6):4773–83.
- [133] Liu W-L, Gong Y-J, Chen W-N, Liu Z, Wang H, Zhang J. Coordinated charging scheduling of electric vehicles: a mixed-variable differential evolution approach. *IEEE Trans Intell Transp Syst* 2020;21(12):5094–109. <https://doi.org/10.1109/TITS.2019.2948596>.
- [134] Zheng Y, Song Y, Hill DJ, Meng K. Online distributed MPC-based optimal scheduling for EV charging stations in distribution systems. *IEEE Trans Ind Inf* 2018;15(2):638–49.
- [135] Chekired DA, Khoukhi L, Mouftah HT. Decentralized cloud-SDN architecture in smart grid: a dynamic pricing model. *IEEE Trans Ind Inf* 2017;14(3):1220–31.
- [136] Wei Z, Li Y, Zhang Y, Cai L. Intelligent parking garage EV charging scheduling considering battery charging characteristic. *IEEE Trans Ind Electron* 2017;65(3): 2806–16.
- [137] Li Y, Hu B. A consortium blockchain-enabled secure and privacy-preserving optimized charging and discharging trading scheme for electric vehicles. *IEEE Trans Intell Transp Syst* 2021;17(3):1968–77. <https://doi.org/10.1109/TII.2020.2990732>.
- [138] Varshosaz F, Moazzami M, Fani B, Siano P. Day-ahead capacity estimation and power management of a charging station based on Queuing theory. *IEEE Trans Ind Inf* 2019;15(10):5561–74.
- [139] Zhou B, et al. Optimal coordination of electric vehicles for virtual power plants with dynamic communication spectrum allocation. *IEEE Trans Intell Transp Syst* 2021;17(1):450–62. <https://doi.org/10.1109/TII.2020.2986883>.
- [140] Chekired DA, Khoukhi L, Mouftah HT. Fog-computing-based energy storage in smart grid: a Cut-Off priority Queuing model for plug-in electrified vehicle charging. *IEEE Trans Ind Inf* 2019;16(5):3470–82.
- [141] Gupta V, Konda SR, Kumar R, Panigrahi BK. Multiaggregator collaborative electric vehicle charge scheduling under variable energy purchase and EV cancellation events. *IEEE Trans Ind Inf* 2017;14(7):2894–902.
- [142] Alsabbagh A, Yin H, Ma C. Distributed electric vehicles charging management with social contribution concept. *IEEE Trans Ind Inf* 2019;16(5):3483–92.
- [143] Song L, Wang J, Yang D. Optimal placement of electric vehicle charging stations based on Voronoi diagram. In: 2015 IEEE international Conference on information and automation. IEEE; 2015. p. 2807–12.
- [144] Logenthiran T, Srinivasan D, Tan Zong S. Demand side management in smart grid using Heuristic optimization. *IEEE Trans Smart Grid* 2012;3(3):1244–52. <https://doi.org/10.1109/TSG.2012.2195686>.
- [145] Saeed MA, et al. Energy management system in smart buildings based coalition game theory with fog platform and smart meter infrastructure. *Sci Rep* 2023;13(1). <https://doi.org/10.1038/s41598-023-29209-4>. 2023-17.
- [146] Zeifman M, Roth K. Nonintrusive appliance load monitoring: review and outlook. *IEEE Trans Consum Electron* 2011;57(1):76–84. <https://doi.org/10.1109/TCE.2011.5735484>.
- [147] Kelly J, Knottenbelt W. Neural NILM: deep neural networks applied to energy disaggregation. New York, NY, USA: ACM; 2015. p. 55–64. <https://doi.org/10.1145/2821650.2821672>.
- [148] Rafiq H, Manandhar P, Rodriguez-Ubinas E, Ahmed Qureshi O, Palpanas T. A review of current methods and challenges of advanced deep learning-based non-intrusive load monitoring (NILM) in residential context. *Energy Build* 2024; 305:113890. <https://doi.org/10.1016/j.enbuild.2024.113890>.
- [149] Diamantoulakis PD, Kapinas VM, Karagiannidis GK. Big data analytics for dynamic energy management in smart grids. *Big data research* 2015;2(3):94–101. <https://doi.org/10.1016/j.bdr.2015.03.003>.
- [150] Chen S, Zhang G, Xia X, Chen Y, Setunge S, Shi L. The impacts of occupant behavior on building energy consumption: a review. *Sustain Energy Technol Assessments* 2021;45:101212. <https://doi.org/10.1016/j.seta.2021.101212>.
- [151] Mocanu E, Nguyen PH, Gibescu N, Kling WL. Deep learning for estimating building energy consumption. *Sustain Energy Grids Netw* 2016;6:91–9. <https://doi.org/10.1016/j.segan.2016.02.005>.
- [152] Zhang L, Huo M, Zhou T, Pan J, Xu Y. Energy flexibility Realization in grid-Interactive buildings for demand response: state-of-the-art review on strategies, resources, control, and KPIs. *BASEL: Mdpi* 2025;18:4960.
- [153] Wang R, Qiu H, Gao H, Li C, Dong ZY, Liu J. Adaptive Horizontal federated learning-based demand response baseline load estimation. *IEEE Trans Smart Grid* 2024;15(2):1659–69. <https://doi.org/10.1109/TSG.2023.3318418>.
- [154] Lin J, Ma J, Zhu J. A privacy-preserving federated learning method for probabilistic community-level behind-the-meter solar generation disaggregation. *IEEE Trans Smart Grid* 2022;13(1):268–79. <https://doi.org/10.1109/TSG.2021.3115904>.
- [155] Wang Y, Chen Q, Hong T, Kang C. Review of smart meter data analytics: applications, methodologies, and challenges. *IEEE Trans Smart Grid* 2019;10(3): 3125–48. <https://doi.org/10.1109/TSG.2018.2818167>.
- [156] Ahmadi H, Marti JR. Load decomposition at smart meters level using Eigenloads approach. *IEEE Trans Power Syst* 2015;30(6):3425–36. <https://doi.org/10.1109/TPWRS.2014.2388193>.
- [157] Ouzine J, Marzouq M, Dosse Bennani S, Lahrech K, El Fadili H. New hybrid deep learning models for multi-target NILM disaggregation. *Energy Effic* 2023;16(7): 82. <https://doi.org/10.1007/s12053-023-10161-1>.
- [158] Cimen H, Wu Y, Wu Y, Terriche Y, Vasquez JC, Guerrero JM. Deep learning-based probabilistic Autoencoder for residential energy disaggregation: an adversarial approach. *IEEE Trans Ind Inf* 2022;18(12):8399–408. <https://doi.org/10.1109/TII.2022.3150334>.
- [159] Liang H, Ma J. Separation of residential space Cooling usage from smart meter data. *IEEE Trans Smart Grid* 2020;11(4):3107–18. <https://doi.org/10.1109/TSG.2020.2965958>.
- [160] Petropoulos F, Kourentzes N, Assimakopoulos V. Cross-temporal aggregation: improving the forecast accuracy of hierarchical electricity consumption. *Appl Energy* 2020;261. <https://doi.org/10.1016/j.apenergy.2019.114339>.
- [161] Iyengar S, Lee S, Irwin D, Shenoy P. Analyzing energy usage on a City-scale using utility smart meters. Presented at the Proceedings of the 3rd ACM international Conference on systems for energy-efficient built environments, Palo Alto, CA, USA. 2016. <https://doi.org/10.1145/2993422.2993425> [Online]. Available:.
- [162] Almassalkhi M, et al. Hierarchical, grid-aware, and Economically optimal coordination of distributed energy resources in realistic distribution systems. *Energies* 2022;13(23):6399. <https://doi.org/10.3390/en13236399>.
- [163] Wang X, Wang H, Bhandari B, Cheng L. AI-empowered methods for smart energy consumption: a review of load forecasting, anomaly detection and demand response. *International journal of precision engineering and manufacturing-green Technology* 2024;11(3):963–93. <https://doi.org/10.1007/s40684-023-00537-0>.
- [164] Gebhard T, Tundis A, Steinke F. Explainable anomaly detection for grid monitoring using probabilistic load forecasting. 2025 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm) 2025;29:1–7. <https://doi.org/10.1109/SmartGridComm65349.2025.11204576>.
- [165] Wang X, Zhao T, Liu H, He R. Power consumption predicting and anomaly detection based on long short-term memory neural network. In: 2019 IEEE 4th international Conference on cloud computing and big data analysis (ICCCBDA), 12–15 April 2019; 2019. p. 487–91. <https://doi.org/10.1109/ICCCBDA.2019.8725704>.
- [166] Park S, Yoon S, Lee B, Ko S, Hwang E. Probabilistic forecasting based joint detection and imputation of clustered Bad data in residential electricity loads. *Energies* 2021;14(1):165. <https://doi.org/10.3390/en14010165>.
- [167] Hossen, Md Sabbir and Ramasamy, Gobbi and Eng, Ngu Eng and Sarker, Md Tanjil and Nabi, Md Serajun, Artificial intelligence for sustainable EV charging: integrated forecasting, anomaly detection, and scheduling in smart grids. Available at SSRN: <https://ssrn.com/abstract=5432752orhttp://dx.doi.org/10.2139/ssrn.5432752>.
- [168] Yang T, Li Y, Zeng Y. Secure state of charge estimation for lithium-ion batteries under deception attacks based on attack-resilient fractional-order extended Kalman filter. *J Energy Storage* 2024;95:112438. <https://doi.org/10.1016/j.est.2024.112438>.
- [169] Zhao H, Yan X, Ren H. Quantifying flexibility of residential electric vehicle charging loads using non-intrusive load extracting algorithm in demand response. *Sustain Cities Soc* 2019;10/01/2019;50:101664. <https://doi.org/10.1016/j.scs.2019.101664>.
- [170] Park H. Human comfort-based-home energy management for demand response participation. *Energies* 2020;13(10):2463. <https://doi.org/10.3390/en13102463>.

- [171] Wang X, Yao Z, Papaefthymiou M. A real-time electrical load forecasting and unsupervised anomaly detection framework. *Appl Energy* 2023;330:120279. <https://doi.org/10.1016/j.apenergy.2022.120279>.
- [172] Wang R, Rayhana R, Gholami M, Herrera OE, Liu Z, Mérida W. Multi-task deep learning for large-scale buildings energy management. *Energy Build* 2024;03/15/2024;307:113964. <https://doi.org/10.1016/j.enbuild.2024.113964>.
- [173] S. Dawn et al., "Enhancing grid stability and sustainability in electrical markets: a review on the synergy of renewable energy and electric vehicles," *Energy Explor Exploit*, vol. 0, no. 0, p. 01445987251383177, doi: 10.1177/01445987251383177.
- [174] Chaudhari AY, Mulay P, Chavan S. The role of smart electricity meter data analysis in driving sustainable development. *MethodsX* 2025;06/01/2025;14:103196. <https://doi.org/10.1016/j.mex.2025.103196>.
- [175] Matthew B, Felix J. Load forecasting techniques in energy systems. 2025.
- [176] Nowcasting Solar PV. Project No. NIA2_NGESO002, national energy system operator, ENA smarter networks portal. https://smarter.energynetworks.org/projects/nia2_ngeso002/. Accessed: Nov. 2025.
- [177] Alboom RK. Energy forecasting inaccuracies and their direct impact on grid performance. *ProQuest Dissertations & Theses*; 2025.
- [178] Li Z, Cao X. Analysis of information feedback on residential energy conservation and the implications: the case of China. *Front Environ Sci* 2021;9. <https://doi.org/10.3389/fenvs.2021.626890>.
- [179] Asmare F, Jaraitė J, Kazukauskas A. The effect of descriptive information provision on electricity consumption: Experimental evidence from Lithuania. *Energy Econ* 2021;104:105687. <https://doi.org/10.1016/j.eneco.2021.105687>.
- [180] Mohammadmoradi H, Gnawali O, Moss D, Boelzle R, Wang G. Effectiveness of a task-based residential energy efficiency program in Oahu. In: 2017 sustainable Internet and ICT for sustainability (SustainIT), 6-7 Dec. 2017; 2017. p. 1–8. <https://doi.org/10.23919/SustainIT.2017.8379792>.
- [181] Micheli G, Soda E, Vespucci MT, Gobbi M, Bertani A. Big data analytics: an aid to detection of non-technical losses in power utilities. *Comput Manag Sci* 2019;16(1–2):329–43. <https://doi.org/10.1007/s10287-018-0325-x>.
- [182] Balaras CA, Dascalaki EG, Patsioti M, Droutsas KG, Kontoyiannidis S, Cholewa T. Carbon and Greenhouse Gas emissions from electricity consumption in European union buildings. *Buildings* 2024;14(1):71. <https://doi.org/10.3390/buildings14010071>.
- [183] Wieberneit F, Crisostomi E, Quinn A, Shorten R. Predictive control of EV Overnight charging with multi-Session flexibility. 2025. <https://doi.org/10.48550/arxiv.2505.05087>.
- [184] Anwar MB, et al. Assessing the value of electric vehicle managed charging: a review of methodologies and results. *Energy Environ Sci* 2022;15(2):466–98. <https://doi.org/10.1039/d1ee02206g>.
- [185] Khan Y, Ur Rehman A, Sardar A, Wadud Z, Ashfaq M. Optimizing energy consumption for smart home using machine learning techniques. 2025.
- [186] Oracle. Australia's United Energy mines deep ADMS value – customer Success/ Australia's United Energy mines deep ADMS value, Oracle and/or its affiliates, PDF case study [Online]. Available: <https://www.oracle.com/a/ocom/docs/in dustries/utilities/united-energy-case-study.pdf>; June 9, 2022.
- [187] Clark E. The effect of smart grid technology on energy consumption management in Canada. *International Journal of Technology and Systems* 2024;9(3):38–50. <https://doi.org/10.47604/ijts.2815>.
- [188] Tao F, Zhang H, Liu A, Nee AYC. Digital twin in industry: state-of-the-art. *IEEE Trans Ind Inf* 2019;15(4):2405–15. <https://doi.org/10.1109/TII.2018.2873186>.
- [189] Jiang Z, Lv H, Li Y, Guo Y. A novel application architecture of digital twin in smart grid. *J Ambient Intell Hum Comput* 2022;13(8):3819–35. <https://doi.org/10.1007/s12652-021-03329-z>.
- [190] Yang W, Xiang W, Yang Y, Cheng P. Optimizing federated learning with deep Reinforcement learning for digital twin empowered industrial IoT. *IEEE Trans Ind Inf* 2023;19(2):1884–93. <https://doi.org/10.1109/TII.2022.3183465>.
- [191] Chekired DA, Khoukhi L, Moufah HT. Fog-computing-based energy storage in smart grid: a Cut-Off priority Queuing model for plug-in electrified vehicle charging. *IEEE Trans Ind Inf* 2020;16(5):3470–82. <https://doi.org/10.1109/TII.2019.2940410>.
- [192] Zhang H, Qin B, Tu T, Guo Z, Gao F, Wen Q. An adaptive encryption-as-a-service architecture based on fog computing for real-time substation communications. *IEEE Trans Ind Inf* 2020;16(1):658–68. <https://doi.org/10.1109/TII.2019.2948113>.
- [193] Sharma A, Sangal AL, Sharma KC. Privacy-preserving prosumer profiling using smart meter data. *Sustain Energy Grids Netw* 2025;09/01/2025;43:101784. <https://doi.org/10.1016/j.segan.2025.101784>.
- [194] Bibi H, Abolhasan M, Lipman J, Abdollahi M, Ni W. A comprehensive survey on privacy-preserving technologies for Smart Grids. *Comput Electr Eng* 2025;124:110371. <https://doi.org/10.1016/j.compeleceng.2025.110371>.
- [195] Voyez A, Allard T, Avoine G, Cauchois P, Fromont E, Simonin M. The privacy cost of fine-grained electrical consumption data. *Sci Rep* 2025;15(1):17391. <https://doi.org/10.1038/s41598-024-78285-7>.
- [196] National Institute of Standards and Technology. NISTIR 7628 Revision 1: smart grid cybersecurity guidelines, section 5.3 – privacy, identification and protection of smart grid information. Gaithersburg, MD, USA, Jun: U.S. Dep't of Commerce; 2014. <https://nvlpubs.nist.gov/nistpubs/ir/2014/nist.ir.7628r1.pdf>. Accessed: Oct. 2025.
- [197] ISO/IEC 27557. Privacy risk management — governance of personal data. Geneva, Switzerland: International Organization for Standardization; 2022. <https://www.iso.org/standard/71675.html>. Accessed: Oct. 2025.
- [198] Ge M, Chren S, Rossi B, Pitner T, Corchuelo R, Abramowicz W. Data quality management framework for smart grid systems. In: *Lecture Notes in Business information processing*. vol. 354. Switzerland: Springer International Publishing AG; 2019. p. 299–310.
- [199] Fernández JD, Menci SP, Lee CM, Rieger A, Fridgen G. Privacy-preserving federated learning for residential short-term load forecasting. *Appl Energy* 2022/11/15/2022;326:119915. <https://doi.org/10.1016/j.apenergy.2022.119915>.
- [200] Wen M, Xie R, Lu K, Wang L, Zhang K. FedDetect: a novel privacy-preserving federated learning framework for energy theft detection in smart grid. *IEEE Internet Things J* 2022;9(8):6069–80. <https://doi.org/10.1109/JIOT.2021.3110784>.
- [201] Eibl G, Engel D. Differential privacy for real smart metering data. *Comput Sci Res Dev* 2017/03/01 2017;32(1):173–82. <https://doi.org/10.1007/s00450-016-0310-y>.
- [202] Ali M, Suchismita M, Ali SS, Choi BJ. Privacy-preserving machine learning for IoT-integrated smart grids: recent advances, opportunities, and challenges. *BASEL: Mdpi* 2025;18:2515.
- [203] Marandi A, Alves PGMR, Aranha DF, Jacobsen RH. Lattice-based homomorphic encryption for privacy-preserving smart meter data analytics. *Comput J* 2024;67(5):1687–98. <https://doi.org/10.1093/comjnl/bxad093>.
- [204] Ghalib M, Ahmed A, Alshiab I, Bouida Z, Ibnkahla M. Implementation of a smart grid communication system compliant with IEEE 2030, vol. 5; 2018.
- [205] Ngcobo T, Ghayoor F. An overview of DLMS/COSEM and g3-plc for smart metering applications. *Int J Smart Sens Intell Syst* 2022;15:20220011. <https://doi.org/10.2478/ijssis-2022-0011>.
- [206] Kim HJ, et al. A comprehensive review of practical issues for interoperability using the common information model in smart grids. *Energies* 2020;13(6):1435. <https://doi.org/10.3390/en13061435>.
- [207] Leszczyna R. Standards with cybersecurity controls for smart grid—a systematic analysis. *Int J Commun Syst* 2019;32(6):n/a. <https://doi.org/10.1002/dac.3910>.
- [208] A. Gupta, A., 2025. Exploring edge computing and cloud computing: a comparative study of features and applications. *Int J Appl Math*, 38(6s), pp.994–1018. doi: 10.12732/ijam.v38i6s.448.
- [209] Galij S, Pawlak G, Grzyb S. Modeling data sovereignty in public cloud—a comparison of existing solutions. *Appl Sci* 2024;14(23):10803. <https://doi.org/10.3390/app142310803>.
- [210] Bassi C, Souza SNA. Challenges to implementing effective data governance: a literature review. 2023. p. 17–28.
- [211] Bernardo BMV, Mamede HS, Barroso JMP, dos Santos VMPD. Data governance & quality management—Innovation and breakthroughs across different fields. *J Innov Knowl* 2024/10/01/2024;9(4):100598. <https://doi.org/10.1016/j.jik.2024.100598>.